

Benchmarking trigonometric continuous-variable gate primitives with trapped ions

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Hybrid continuous-discrete-variable quantum processors can represent bosonic degrees of freedom directly in oscillator modes, or qumodes, while using qubits for control, readout, and nonlinear operations. Recently proposed trigonometric continuous-variable (CV) gate sets promote periodic functions of oscillator quadratures to elementary operations, making them natural primitives for compact variables, rotor models, lattice gauge theories, and anharmonic dynamics. Here we experimentally demonstrate and benchmark one-qumode cosine gates, and perform a mode-resolved marginal benchmark of two-qumode cosine-gate implementations, on the QSCOUT trapped-ion quantum platform. Our implementation uses collective motional modes of three- and four-ion $^{171}\text{Yb}^+$ chains and realizes finite-step trigonometric-gate circuits through hybrid qubit-qumode operations and conditional phase-space displacements. In contrast to previous theoretical and compilation work, we focus on the gate-level characterization of the trigonometric primitives. We measure Fock-space transition probabilities, study their dependence on gate parameters and Trotter step number, and compare with simulations incorporating thermal initialization and motional dephasing. We also derive ideal gate matrix elements and phase-space diagnostics, connecting the measurements to the non-Gaussian structure generated by these gates. These results establish trigonometric CV gates as reusable building blocks for bosonic Hamiltonian simulations and hybrid quantum algorithms requiring intrinsically non-polynomial operations.

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I. INTRODUCTION

Quantum architectures that combine discrete-variable (DV) qubits with continuous-variable (CV) bosonic oscillators or qumodes provide a flexible route toward quantum information processing beyond purely qubit-based platforms [1–3]. Hybrid qubit-qumode registers are naturally realized in trapped ions, superconducting circuits, and cavity-QED systems [4–13], where long-lived oscillator modes can be coherently controlled through ancillary qubits. The resulting hardware natively supports entangling qubit-qumode interactions, conditional displacements, sideband couplings, spin-dependent forces [14], programmable phononic networks [15], and non-Gaussian resources [16, 17], making it well-suited for both near-term algorithmic demonstrations and long-term fault-tolerant architectures.

The ability to directly use bosonic degrees of freedom is particularly valuable for problems whose natural Hilbert space is infinite-dimensional. Rather than encoding oscillator states into a qubit register, CV-DV architectures can represent bosonic modes directly and use qubits to control and read out the modes, and to effect nonlinear operations [18, 19]. This perspective has motivated a growing body of work on hybrid qubit-qumode instruction sets, compilation strategies, and applications to quantum simulation, quantum error correction, and quantum algorithms [7, 12, 20–22]. For example, in quantum error correction, bosonic encodings such as GKP, cat, and binomial codes provide hardware-efficient ways of storing logical information in oscillator modes [9, 23–25], and logical qubit rotations can be achieved using the interaction with qubits. These ideas have progressed from theoretical proposals to experimental realizations across several physical platforms [10, 26–28].

Hybrid qubit-qumode systems are also naturally aligned with the structure of many quantum simulation tasks. In lattice field theories and related lattice models, fermionic or spin degrees of freedom can be mapped to qubits [29, 30], while gauge fields, scalar fields, phonons [31], or other bosonic degrees of freedom can be represented by qumodes [32–55]. This division of labor can reduce encoding overhead and provides a hardware-native route to simulating models with interacting discrete and continuous degrees of freedom [56]. Beyond lattice models, qumodes are also naturally suited to molecular and vibra-

tional quantum dynamics, where the relevant degrees of freedom are harmonic or anharmonic vibrational modes [57]. A key example is the simulation of vibronic spectra, including Franck-Condon spectra, where transition amplitudes between vibrational wavefunctions can be formulated directly in terms of bosonic modes and Gaussian/non-Gaussian CV operations [58–61]. These examples illustrate that hybrid qubit-qumode processors provide a natural language for problems involving both discrete electronic structure and continuous bosonic dynamics.

A central question for such architectures is which elementary CV and hybrid gates should be employed as natural building blocks. Typically, CV universality is formulated in terms of finite polynomial functions of the canonical quadratures $\hat{x}$ and $\hat{p}$. Gaussian operations, generated by Hamiltonians at most quadratic in the quadratures, together with a suitable non-Gaussian resource, provide a universal gate set [1]. However, various physical systems of interest involve periodic structures that are not naturally represented by low-order polynomials. Examples include lattice gauge theories with compact gauge variables, rotor models, sine-Gordon-type interactions [62], and more general periodic potentials. In such cases, polynomial approximations may require high orders or large circuit depths to faithfully represent the relevant operators over the physically occupied region of phase-space.

In this work, we build on the trigonometric continuous-variable gate framework introduced in Ref. [63] and independently developed in Ref. [64]. These works established trigonometric gates as theoretical and compilation primitives for hybrid CV-DV quantum information processing. In particular, Ref. [63] introduced a Fourier-like universality paradigm based on trigonometric functions of qumode quadratures and applied it to hybrid simulations of the sine-Gordon model, while Ref. [64] developed a programmable Fourier-synthesis approach for generating anharmonic single-mode potentials and nonlinear multimode couplings. The present work has a different and complementary focus. Rather than using trigonometric gates as components of a higher-level simulation algorithm, we isolate the elementary cosine gates themselves and perform a gate-level experimental and analytical characterization of their action on trapped-ion motional modes.

The basic gates studied here are the one- and two-qumode operations

$$e^{-i\theta \cos(c\hat{x})}, \quad e^{-i\theta \cos(c_1\hat{x}_1 + c_2\hat{x}_2)}, \quad (1)$$

together with their finite-step circuit approximations using hybrid qubit-qumode interactions. Our contribution is threefold. First, we experimentally implement these elementary trigonometric CV gates on the Quantum Scientific Computing Open User Testbed (QSCOUT), using collective motional modes of trapped $^{171}\text{Yb}^+$ ion chains [65] as qumodes and hyperfine states as ancillary qubits. Second, we characterize the gates at the level of their Fock-space transition probabilities [66], including the dependence on the gate parameters, the number of Trotter steps, postselection, thermal initialization, and motional dephasing. Third, we derive analytical expressions for the ideal gate matrix elements, characteristic functions, Wigner functions, and Wigner negativity, thereby connecting the experimentally measured observables to the non-Gaussian phase-space structure generated by trigonometric gates.

Recent experimental work has demonstrated related CV-DV techniques for programmable simulations of anharmonic dynamics in trapped ions [67]. Related trapped-ion experiments have also shown that hybrid spin-motion interactions can generate effective nonlinear bosonic operations beyond Gaussian control, including squeezing, trisqueezing, and quadsqueezing [68]. Our work is complementary to that demonstration. Whereas Ref. [67] uses related synthesis ideas to realize tunable anharmonic potentials and observe wavepacket dynamics in specific target Hamiltonians, here we benchmark the underlying trigonometric gate primitives directly. In particular, we study both single-mode and two-mode cosine gates, compare finite-Trotter implementations against the ideal trigonometric evolution, and analyze how experimental imperfections modify the measured transition probabilities. This gate-level perspective provides a quantitative foundation for using trigonometric operations as reusable building blocks in future simulations of periodic bosonic Hamiltonians, compact gauge fields, rotor models, and hybrid quantum algorithms requiring intrinsically non-polynomial CV structure.

The remainder of this paper is organized as follows. In Section II, we outline the foundations of CV-DV universality and explain how trigonometric gates provide a natural non-polynomial complement to the usual polynomial gate paradigm. In Section III, we introduce the trapped-ion implementation on QSCOUT. In Section IV, we analyze ideal one- and two-qumode cosine gates, including their Fock-basis matrix elements, characteristic functions,

Wigner functions, and Wigner negativity. In Section V, we describe the finite-step circuit realization of these gates using conditional displacements and ancillary qubits. In Section VI, we present the experimental measurements and compare them with analytical and open-system simulations. In Section VII, we summarize the results and discuss future applications.

II. UNIVERSALITY AND ELEMENTARY GATES

The universality of CV quantum computing is normally formulated via a finite polynomial expansion in terms of the canonical quadratures $\hat{x}$ and $\hat{p}$ [1, 69]. The goal is to approximate the general N -qumode bosonic unitary $e^{i\theta f(\hat{\mathbf{x}}, \hat{\mathbf{p}})}$ with f an arbitrary finite polynomial of the quadratures. We introduce the following compact notation for the phase-space vector of N qumodes $\hat{\mathbf{Q}} = (\hat{x}_1, \dots, \hat{x}_N, \hat{p}_1, \dots, \hat{p}_N)^T$. We can then express the bosonic contribution as any N -modes finite degree- d polynomial

$$f(\hat{\mathbf{Q}}) = \sum_{k=1}^d \sum_{i_1, \dots, i_k=1}^{2N} c_{i_1, \dots, i_k} \hat{Q}_{i_1} \dots \hat{Q}_{i_k}, \quad (2)$$

with coefficients $c_{i_1, \dots, i_k} \in \mathbb{R}$.

An extension of this notion of universality to hybrid CV-DV gates allows for the simulation of unitaries where the Hermitian exponent is the tensor product $f(\hat{\mathbf{x}}, \hat{\mathbf{p}}) \otimes P_n$, with $P_n \in \{\mathbb{I}, X, Y, Z\}^{\otimes n}$ a Pauli string acting on n qubits. This can be realized using the compact elementary gate set [69–71]

$$\{\text{CD}_{\mathbf{n}\cdot\boldsymbol{\sigma}}(\alpha), R_{\mathbf{n}\cdot\boldsymbol{\sigma}}(\theta), \text{BS}(\alpha)\}, \quad (3)$$

where the conditional displacement, the qubit rotation, and the qumode beam splitter are given by

$$\begin{aligned} \text{CD}_{\mathbf{n}\cdot\boldsymbol{\sigma}}(\alpha) &= e^{(\alpha \hat{a}^\dagger - \alpha^* \hat{a}) \otimes \mathbf{n}\cdot\boldsymbol{\sigma}}, \\ R_{\mathbf{n}\cdot\boldsymbol{\sigma}}(\theta) &= e^{-i\frac{\theta}{2} \mathbf{n}\cdot\boldsymbol{\sigma}}, \\ \text{BS}(\alpha) &= e^{\alpha \hat{a}_1 \hat{a}_2^\dagger - \alpha^* \hat{a}_1^\dagger \hat{a}_2}. \end{aligned} \quad (4)$$

Here, $\mathbf{n}$ denotes a Bloch vector and $\boldsymbol{\sigma} = (X, Y, Z)$, and $\alpha = \theta e^{i\phi}$. For example, any higher power of the Hermitian generator $i(\alpha \hat{a}^\dagger - \alpha^* \hat{a})$ of the displacements can be realized by concatenating CD gates with different control Pauli operators. The same protocol applies to the generation of higher powers of the non-polynomial trigonometric gates, discussed in the following.

We note that, in practice, hardware implementations rarely restrict themselves to the minimal universal gate set, since this would neglect operations naturally supported by the underlying physical platform. For example, with trapped ion platforms, the red and blue sideband gates can be natively realized, which are given by

$$\begin{aligned}\text{RSB}(z) &= e^{i(z\hat{a}\sigma^+ + z^*\hat{a}^\dagger\sigma^-)}, \\ \text{BSB}(z) &= e^{i(z\hat{a}\sigma^- + z^*\hat{a}^\dagger\sigma^+)}.\end{aligned}\quad (5)$$

For example, the blue sideband gate is used for the qumode readout. These two unitaries are also referred to as the Jaynes-Cummings (JC) and Anti-Jaynes-Cummings (AJC) gates, respectively. See Ref. [72] for more information about the gate set available with QSCOUT.

Due to their non-polynomial nature, the trigonometric gates introduced in Refs. [63, 64] go beyond the notions of universality discussed above. In addition, the previously discussed notion of universality is intrinsically local in phase-space because it corresponds to a Taylor-like representation of the Hermitian generator. Consequently, approximating operators with global or periodic structures may require high-order polynomials and large circuit depths. The non-polynomial trigonometric gates permit the construction of bosonic unitaries whose Hermitian exponent is a finite degree- d polynomial, not of the quadratures themselves, but of their sines and cosines, yielding the Fourier-like representation of the general bosonic unitary $e^{i\theta g(\hat{x}, \hat{p})}$, where

$$g(\hat{\mathbf{Q}}) = \sum_{\substack{i,j \\ i+j \leq d}} c_{ij} \cos^i(f_c(\hat{\mathbf{Q}})) \sin^j(f_s(\hat{\mathbf{Q}})), \quad (6)$$

with f_c and f_s two arbitrary commuting polynomials of the quadratures and $c_{ij} \in \mathbb{R}$. These non-polynomial gates are naturally synthesized on hybrid CV-DV platforms as conditional trigonometric gates. The simplest example is

$$e^{i\theta \cos(\hat{A}) \otimes \sigma_i}, \quad (7)$$

with $\hat{A}$ any bosonic Hermitian operator. This framework naturally enables the generation of higher-power trigonometric terms by exploiting the Lie algebraic structure of the qubit operators, just like in standard CV-DV universality. With the same logic, gates that mix quadrature powers and trigonometric powers are also possible. Namely,

$$e^{i\theta f(\hat{\mathbf{Q}})g(\hat{\mathbf{Q}}) \otimes \sigma_j}, \quad (8)$$

where we use the same notation as in Eqs. (2) and (6).

III. SIMULATION WITH TRAPPED IONS

In this section, we detail the experimental configuration and the specific parameters of the trapped-ion platform utilized to implement the trigonometric gates.

The experiments were performed in a small register of trapped $^{171}\text{Yb}^+$ ions in the Quantum Scientific Computing Open User Testbed at Sandia National Laboratories [65]. An N -ion chain consists of up to $3N$ motional degrees of freedom, N eigenmodes in the axial direction, and N in each of the two orthogonal radial directions, or principal axes. We address these motional modes through motionally-sensitive Raman transitions driven in a counter-propagating configuration. In this device, those beams are orthogonal to the axial direction, only allowing access to the two radial manifolds of motional modes. Additionally, the surface electrode trap provides the ability to rotate the radial principal axes [73], and so we rotate the axes such that the lower-frequency manifold of radial modes is more strongly coupled to the laser drives. All of the experiments are thus performed on motional eigenmodes within this lower-frequency manifold with either $N = 3$ or $N = 4$ ions.

The qubit states of $^{171}\text{Yb}^+$ correspond to the hyperfine $^2\text{S}_{1/2}$ “clock” transition with a splitting of 12.642 GHz and are driven by a pulsed 355-nm laser via a Raman transition detuned off the excited $^2\text{P}_{1/2}$. We choose the qubits as the following hyperfine states $|\uparrow\rangle := |0\rangle = |^2\text{S}_{1/2}, F=0, m_F=0\rangle$ and $|\downarrow\rangle := |1\rangle = |^2\text{S}_{1/2}, F=1, m_F=0\rangle$. The radial motional modes decorate the qubit transition with first-order red- and blue-sidebands ~ 2.2 MHz detuned from the transition. Driving these red- and blue- transitions independently corresponds to JC and AJC interactions, respectively, see Eq. (5).

In the Lamb-Dicke regime, simultaneous red- and blue-sideband drives addressing ion j and motional mode k realize a spin-dependent force Hamiltonian of the form [74, 75]

$$\begin{aligned}\hat{H}_{\text{SDF}}^{(j,k)}(t) &= \\ \frac{1}{2}\eta_{j,k}\Omega\sigma_{\phi_s}^{(j)} &\left(\hat{a}_k e^{-i(\delta t + \phi_m)} + \hat{a}_k^\dagger e^{i(\delta t + \phi_m)}\right),\end{aligned}\quad (9)$$

where Ω is the carrier transition Rabi frequency, and $\eta_{j,k}$ is the ion j ’s coupling strength to the mode k , i.e.

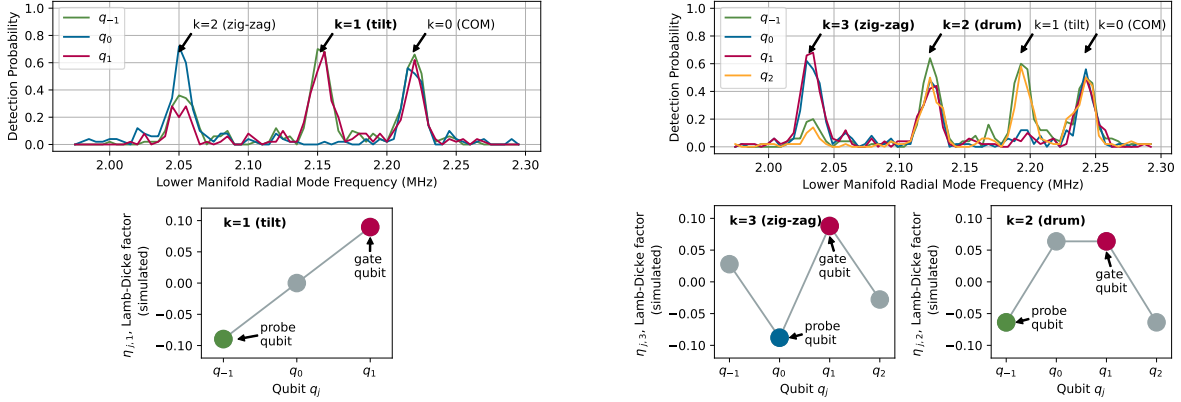


FIG. 1: Trapped ion radial-mode spectroscopy and mode participation. Top row: measured detection probabilities for each qubit as a function of the lower-manifold radial-mode frequency for three-ion (left) and four-ion (right) chains. The resonances identify the collective radial motional modes, labeled by mode index and character. Bottom row: simulated Lamb-Dicke factors $\eta_{j,k}$ for the modes used in the one-qumode (three-ion) experiment, $k = 1$ (tilt), and the two-qumode experiment (four-ion), $k = 3$ (zig-zag) and $k = 2$ (drum). Colored markers indicate the probe and gate qubits used in each implementation, and the gray markers denote the remaining ions.

Lamb-Dicke factor, and for these experiments we set the sideband Rabi frequency for any given ion j and participating mode k to be $\eta_{j,k}\Omega = 2\pi \times 16.66$ kHz. δ is then the detuning from the motional sideband, and ϕ_s and ϕ_m are the spin and motional phases. Note that we utilize natural units where $\hbar = 1$, a convention we maintain throughout the rest of the paper. For near resonant pulses, this interaction results in the conditional displacement $\text{CD}_{\mathbf{n},\sigma}(\alpha)$, see Eq. (4), with α determined by the pulse area, detuning, and optical phase. These $\text{CD}_{\mathbf{n},\sigma}(\alpha)$ gates can then be performed on any selected motional mode, with the magnitude of the displacement proportional to the pulse area and the Lamb-Dicke factor, $\eta_{j,k}$. The phase of the displacement is set by the ϕ_m of the interaction, which in practice is set by the difference in phase between the red- and blue-sideband drives.

In these experiments, we utilize ancillary qubits to perform not only the conditional displacements, but also to perform several indirect methods to resolve aspects of the motional state since there are no direct readouts of the motional state available in trapped-ion systems. For clarity, we will refer to the qubit involved in the conditional displacement operations as the “gate qubit,” and the ones that probe the motional characteristics as the “probe qubits.” One method is to perform a full characteristic function readout through mapping a series of real and

imaginary conditional displacements, measuring the qubit state, and extracting $\langle Z \rangle = P_{|\uparrow\rangle} - P_{|\downarrow\rangle}$ for each displacement [76, 77]. Based on the probe qubit’s initial state, $|\uparrow\rangle$ or $|\downarrow\rangle$, this will either reveal the real or the imaginary part, respectively, of the characteristic function, χ . This method is resource intensive, but it provides both the ability to extract Fock state occupancies and phase relationships. Alternatively, applying a BSB gate of varying duration to the probe qubit can provide information on the Fock state occupancies [78]. Because the coupling of the Fock states increases as $\sqrt{n+1}$ with increasing Fock number n , we can extract occupancies from a fit to a series of sinusoids with frequencies increasing by $\sqrt{n+1}$ [79], see Section VI A for more details. Both of these methodologies are amenable to post-selection, whereby we remove instances in which the gate qubit’s state was flipped during the conditional displacements. For all the data presented here, post-selection success was typically 80 – 90%.

For the experiments described below, we utilize two different ion registers. For the single-mode case of the trigonometric gates, we use a three-ion chain: consisting of the gate qubit coupled to a particular mode, while the probe qubit coupled to the same mode measures the Fock occupancy of that mode, see Fig. 1 (left panels). If we assign the mode indices k with the highest-energy mode $k := 0$ (center-of-mass), then we operate this particular gate on the

$k = 1$ (tilt) mode using one of the two outer ions as the gate qubit and the other as the probe qubit. In the case of the two-mode trigonometric gate, we use a four-ion chain, see Fig. 1 (right panels). Here, we use the $k = 2$ (drum) and $k = 3$ (zig-zag) modes as the modes of choice. In particular, $k = 2$ has equal magnitude participation across all ions, while $k = 3$ is more strongly coupled to the two center ions. The gate qubit for the two-mode gate is thus one of the two center ions. For readout, the other center ion acts as the probe qubit for the $k = 3$ mode, while one of the outer ions is the probe qubit for the $k = 2$ mode.

In both register lengths, all non-center-of-mass modes are sideband cooled to near the vacuum state, with a residual phonon number, $\bar{n} \sim 0.1$. The center-of-mass mode is unused due to less efficient sideband cooling, leading to $\bar{n} \sim 0.5$. Motional dephasing is the predominant source of error, with measured coherence times ranging from 500-700 μs for various modes. While motional heating rates vary from mode to mode, non-center-of-mass modes have rates ≤ 100 quanta/s. For instance, in the four-ion register, the tilt ($k = 1$) and zig-zag ($k = 3$) modes have rates ~ 10 quanta/s, while the drum ($k = 2$) mode heats more rapidly ~ 100 quanta/s.

IV. TRIGONOMETRIC CONTINUOUS-VARIABLE GATES

In this section, we present analytical results for the matrix elements of trigonometric gate primitives and the associated characteristic and Wigner functions. In addition, we quantify the nonclassicality generated by these gates by studying the resulting Wigner negativity or mana.

A. The one- and two-qumode cosine gate

While the argument of the trigonometric gates introduced in [63] can, in principle, be any Hermitian operator $\hat{A}$, see Eq. (6), we limit ourselves in this work to the case where $\hat{A}$ is a linear function of the quadratures. When $\hat{A}$ is linear in the quadratures, i.e. $\hat{A} = \mathbf{c} \cdot \hat{\mathbf{Q}}$ with $\mathbf{c} \in \mathbb{R}^{2N}$, the exact cosine gate can be regarded as an infinite series of displacements, where each is weighted by a Bessel function. The relevant examples for this paper are the simplest cases involving the cosine of one or two position quadratures. In terms of operators, we have

$\hat{A}_{1\text{-mode}} = c\hat{x}$ and $\hat{A}_{2\text{-mode}} = \mathbf{c} \cdot \hat{\mathbf{x}}$, with $\mathbf{c} = (c_1, c_2)$ and $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2)$. The generalization to more than two modes is straightforward.

Starting with the single-qumode case, we consider the trigonometric gate unitary $e^{-i\theta \cos(c\hat{x})}$. After applying the Jacobi-Anger expansion, we find

$$e^{-i\theta \cos(c\hat{x})} = \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) e^{ick\hat{x}} = \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) D\left(ik \frac{c}{\sqrt{2}}\right). \quad (10)$$

Here, J_k are Bessel functions and D are displacement gates analogous to the conditional displacement CD, see Eq. (4). The Fock matrix elements of the displacement operators are well known, see Ref. [80]. Therefore, we can directly obtain the Fock-space transition matrix elements of the one-qumode cosine gate, which are given by

$$\begin{aligned} \langle m | e^{-i\theta \cos(c\hat{x})} | n \rangle &= \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) \langle m | e^{ick\hat{x}} | n \rangle \\ &= \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) e^{-\frac{c^2 k^2}{4}} \sqrt{\frac{\min(m, n)!}{\max(m, n)!}} \\ &\quad \times \left(i \frac{ck}{\sqrt{2}} \right)^{|m-n|} L_{\min(m, n)}^{(|m-n|)} \left(\frac{c^2 k^2}{2} \right), \end{aligned} \quad (11)$$

where $L_N^{(\alpha)}(x)$ are the generalized Laguerre polynomials of degree N with parameter α . Non-vanishing matrix elements exist when m and n have equal parity, as can be seen directly from the coordinate space representation

$$\langle m | e^{-i\theta \cos(c\hat{x})} | n \rangle = \int_{-\infty}^{\infty} dx \psi_m(x) \psi_n(x) e^{-i\theta \cos(cx)}, \quad (12)$$

where

$$\psi_n(x) = \frac{\pi^{-1/4}}{\sqrt{2^n n!}} e^{-\frac{x^2}{2}} H_n(x), \quad (13)$$

is the wave function of the n -th Fock state. From Eq. (11), we also find the survival/transition probabilities

$$\begin{aligned} P_{m \leftarrow n}(\theta, c) &\equiv |\langle m | e^{-i\theta \cos(c\hat{x})} | n \rangle|^2 = \\ &= \sum_{k, k' \in \mathbb{Z}} (-i)^k (i)^{k'} J_k(\theta) J_{k'}(\theta) e^{-\frac{c^2(k^2 + k'^2)}{4}} \frac{\min(m, n)!}{\max(m, n)!} \times \\ &\quad \left(\frac{c^2 k k'}{2} \right)^{|m-n|} L_{\min(m, n)}^{(|m-n|)} \left(\frac{c^2 k^2}{2} \right) L_{\min(m, n)}^{(|m-n|)} \left(\frac{c^2 k'^2}{2} \right). \end{aligned} \quad (14)$$

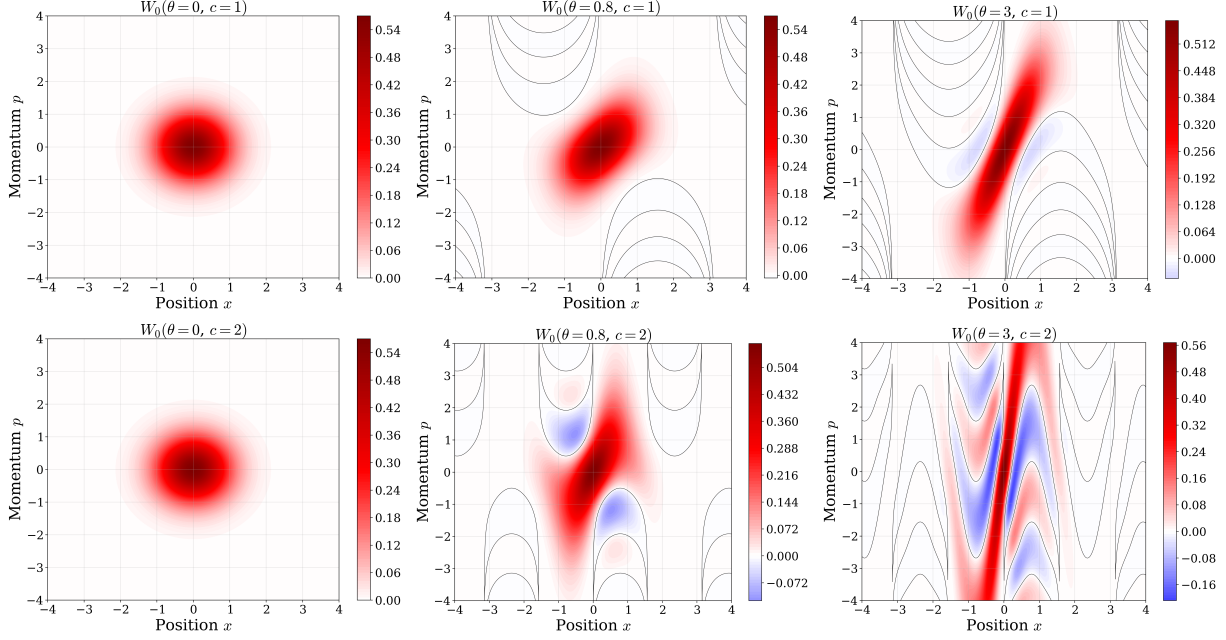


FIG. 2: Wigner function of the single-qumode cosine gate acting on the Fock vacuum state $|0\rangle$ for different values of $\theta = 0, 0.8, 3$ and fixed parameter $c = 1$ (upper row), $c = 2$ (lower row).

For the experimental measurements discussed below, we specifically focus on the case $n = 0$. The calculations simplify considerably because $L_0^{(m)}\left(\frac{c^2 k^2}{2}\right) = 1$. Thus, we obtain

$$P_{m \leftarrow 0}(\theta, c) = \begin{cases} 0, \\ \frac{c^{2m}}{2^m m!} \left| \sum_{k \in \mathbb{Z}} (-1)^k i^k k^m J_k(\theta) e^{-\frac{c^2 k^2}{4}} k^m \right|^2, \end{cases} \quad (15)$$

where the only non-vanishing probabilities are obtained for even m .

Next, we consider the two-qumode extension of the cosine gate. Mirroring the results of the single-mode case, we can once again regard the ideal gate as an infinite sum of displacements

$$e^{-i\theta \cos(\mathbf{c} \cdot \hat{\mathbf{x}})} = \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) e^{i c_1 k \hat{x}_1} e^{i c_2 k \hat{x}_2} = \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) D_1\left(ik \frac{c_1}{\sqrt{2}}\right) D_2\left(ik \frac{c_2}{\sqrt{2}}\right), \quad (16)$$

with each mode getting displaced independently. The matrix elements in the two-mode Fock basis

read

$$\begin{aligned} & \langle m_1 | \langle m_2 | e^{-i\theta \cos(c_1 \hat{x}_1 + c_2 \hat{x}_2)} | n_1 \rangle | n_2 \rangle \\ &= \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) \prod_{j=1}^2 \langle m_j | e^{i c_j k \hat{x}_j} | n_j \rangle \\ &= \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) \prod_{j=1}^2 e^{-\frac{c_j^2 k^2}{4}} \sqrt{\frac{\min(m_j, n_j)!}{\max(m_j, n_j)!}} \\ & \quad \times \left(i \frac{c_j k}{\sqrt{2}} \right)^{|m_j - n_j|} L_{\min(m_j, n_j)}^{(|m_j - n_j|)} \left(\frac{c_j^2 k^2}{2} \right). \end{aligned} \quad (17)$$

Similar to Eq. (15), in this work, we are primarily interested in the Fock vacuum initialization where $|n_1\rangle = |n_2\rangle = |0\rangle$. In this case, the transition probabilities simplify to

$$P_{m_1, 2 \leftarrow 0}(\theta, c) = \left| \sum_{k \in \mathbb{Z}} (-i)^k J_k(\theta) \prod_{j=1}^2 \frac{e^{-\frac{c_j^2 k^2}{4}}}{\sqrt{m_j!}} \left(i \frac{c_j k}{\sqrt{2}} \right)^{m_j} \right|^2. \quad (18)$$

The only non-vanishing probabilities are obtained for $(-1)^{m_1+m_2} = 1$. The total Fock parity must be the same as that of the initial state.

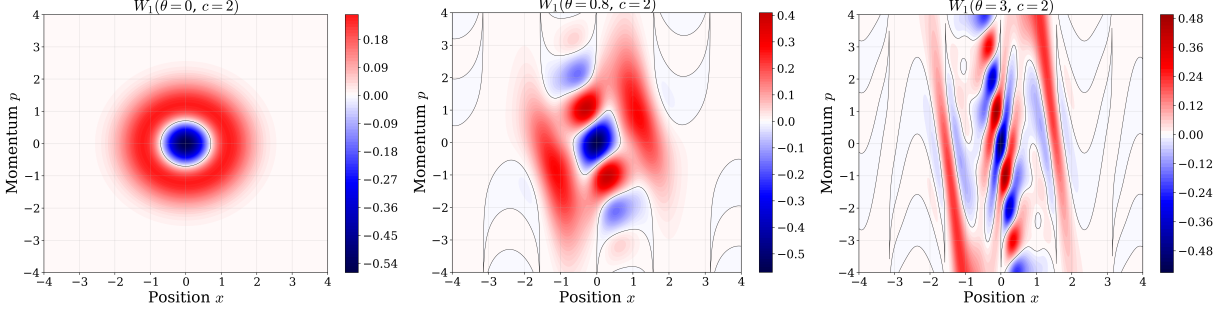


FIG. 3: Wigner function of the single-qumode cosine gate acting on the Fock state $|1\rangle$ for different θ values and fixed parameter $c = 2$.

B. Wigner and characteristic functions

We can also obtain an analytical expression for the Wigner function of a given Fock state evolved under a single-quadrature cosine gate.

We consider the n -th Fock state $|n\rangle$ as the initial state and evolve it with the cosine gate $\exp(-i\theta \cos(c\hat{x}))$. In position space, the evolved wavefunction is therefore $\psi_n(x; \theta, c) = \psi_n(x) e^{-i\theta \cos(cx)}$. The Wigner function is defined as

$$W_n(x, p; \theta, c) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dy e^{ipy} \psi_n^* \left(x + \frac{y}{2}; \theta, c \right) \psi_n \left(x - \frac{y}{2}; \theta, c \right). \quad (19)$$

After some algebra, we find

$$W_n(x, p; \theta, c) = \frac{1}{2\pi} \frac{e^{-x^2}}{2^n n! \sqrt{\pi}} \sum_{k \in \mathbb{Z}} J_k(2\theta \sin(cx)) \times \int dy e^{iy(p-kc/2)} e^{-\frac{y^2}{4}} H_n \left(x - \frac{y}{2} \right) H_n \left(x + \frac{y}{2} \right). \quad (20)$$

For later convenience, we also consider the characteristic function

$$\chi_n(x, p; \theta, c) = \int_{-\infty}^{\infty} du e^{-iux} \psi_n^* \left(u + \frac{p}{2}; \theta, c \right) \psi_n \left(u - \frac{p}{2}; \theta, c \right), \quad (21)$$

which is related to the Wigner function by the Fourier transform

$$\chi_n(x, p; \theta, c) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} du dv e^{-i(ux+vp)} W_n(u, v; \theta, c). \quad (22)$$

Zero Fock state ($n = 0$): Using the Fock ground state $|0\rangle$ as the initial state, we obtain the following result for the Wigner function

$$W_0(x, p; \theta, c) = \frac{e^{-x^2}}{\pi} \sum_{k \in \mathbb{Z}} e^{-(p-kc/2)^2} J_k(2\theta \sin(cx)). \quad (23)$$

In Fig. 2, we show numerical results for the Wigner function for different values of θ and representative choices of the parameter, $c = 1$ and $c = 2$. The non-polynomial evolution produces wave fronts that “collide” with the initial Gaussian peak centered at the origin. For $c = 1$, this collision resembles the evolution of a rotating squeezed state and produces only limited negativity. However, already for $c = 2$, the number of wave fronts encountered by the Gaussian state increases, and the collision rapidly deforms the state, generating large peaks and troughs. This behavior will be quantified more systematically in the next subsection. We note that for an analogous cosine gate involving the momentum quadrature, the wave fronts would approach the central Gaussian peak horizontally rather than vertically.

First excited Fock state ($n = 1$): We obtain

$$W_1(x, p; \theta, c) = (2x^2 + 2p^2 - 1) W_0(x, p; \theta, c) + \frac{e^{-x^2}}{\pi} \sum_{k \in \mathbb{Z}} e^{-(p-\frac{kc}{2})^2} J_k(2\theta \sin(cx)) \left(\frac{k^2 c^2}{2} - 2kcp \right). \quad (24)$$

In Fig. 3, we show representative examples for $c = 2$ and several values of θ . As in the previous case, the action of the cosine gate can be viewed in terms of incoming wave fronts. A key difference is that the initial $n = 1$ Fock state already has a rotationally

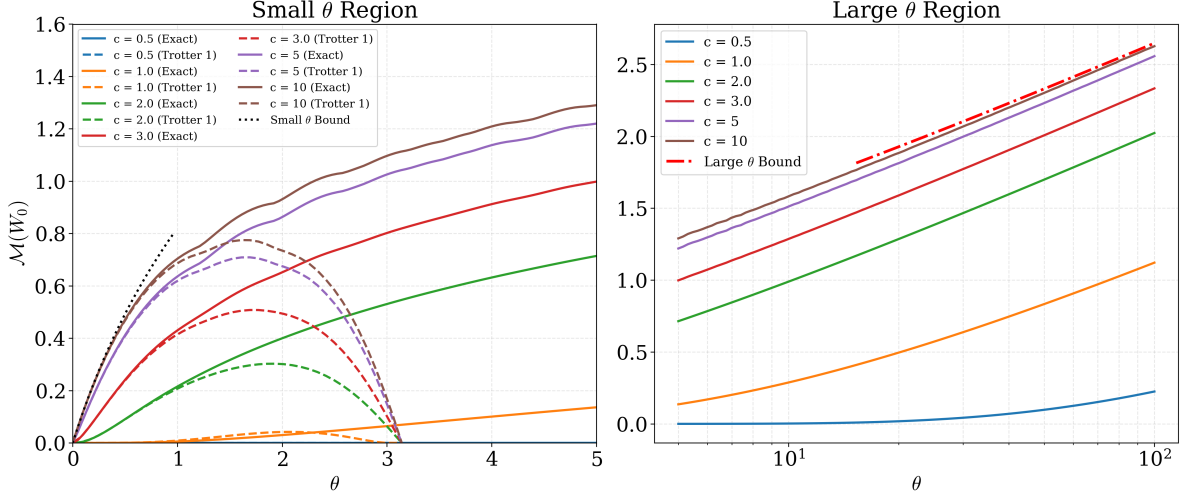


FIG. 4: Evolution of the mana starting in the zero Fock state under the cosine gate evolution as a function of θ for different choices of c . In both panels, we show the bound obtained in the small and large- θ limit, respectively. The left panel compares the Wigner logarithmic negativity obtained using the exact gate with that of its approximation described in Sec. V A.

symmetric Wigner function with regions of negativity.

General Fock state n :

$$W_n(x, p; \theta, c) = \frac{e^{-x^2}}{\pi} \sum_{k \in \mathbb{Z}} e^{-(p - kc/2)^2} J_k(2\theta \sin(cx)) \times (-1)^n L_n[2(x^2 + (p - kc/2)^2)], \quad (25)$$

where $L_n(x)$ are the usual Laguerre polynomials. Its explicit form can be obtained from Eq. (20). Setting $\theta = 0$ or $c = 0$ naturally recovers the well-known Wigner function of the n -th Fock state. Similarly, the characteristic function is found to be

$$\chi_n(x, p; \theta, c) = \sum_{k \in \mathbb{Z}} J_k\left(2\theta \sin\left(\frac{cp}{2}\right)\right) e^{-\frac{(x + kc)^2 + p^2}{4}} \times L_n\left(\frac{(x + kc)^2 + p^2}{2}\right). \quad (26)$$

As a result, even for $c \neq 0$ and $\theta \neq 0$, the state has definite Fock parity and, therefore, satisfies the phase-space self-dual relation

$$\chi_n(x, p; \theta, c) = \pi(-1)^n W_n\left(\frac{p}{2}, -\frac{x}{2}; \theta, c\right). \quad (27)$$

C. Negativity and mana

To quantify the nontrivial action of the cosine gate, we consider its effect on the Fock vacuum $|0\rangle$,

which is the only Gaussian Fock state. The cosine gate generates non-Gaussian features that can be characterized through the Wigner function. In particular, we quantify the resulting nonclassicality by integrating the absolute value of the Wigner function over phase-space.

The mana $\mathcal{M}$, also known in CV systems as the Wigner logarithmic negativity (WLN) [81], is defined as

$$e^{\mathcal{M}(W_0)} = \int dx dp |W_0(x, p; \theta, c)|. \quad (28)$$

Small time evolution ($\theta \ll 1$): Up to order $\mathcal{O}(\theta)$, the only non-trivial contributions come from the Bessel functions J_0 and $J_{\pm 1}$, while all other terms enter at higher orders in θ . After some algebra, we find the following result for the Wigner function

$$W_0(x, p; \theta, c) = \frac{e^{-x^2 - p^2}}{\pi} \left[1 + 2\theta \sin(cx) e^{-c^2/4} \sinh(pc) \right] + \mathcal{O}(\theta^2). \quad (29)$$

Using this expression, the mana can be written as

$$e^{\mathcal{M}(W_0)} \Big|_{1 + \mathcal{O}(\theta)} = 1 + \frac{2\theta}{\sqrt{\pi}} \text{Erf}(c/2) \int_{-\infty}^{\infty} dx e^{-x^2} |\sin(cx)|. \quad (30)$$

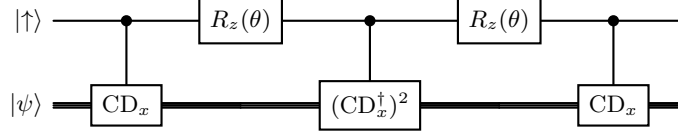


FIG. 5: Circuit that approximates the action of the trigonometric operator $e^{-i\theta \cos(c\hat{x})}$ on a qumode state $|\psi\rangle$ up to order $\mathcal{O}(\theta^2)$ using one ancillary gate qubit. The argument of the conditional displacement is $\text{CD}_x = \text{CD}_x(ic/(2\sqrt{2}))$. Throughout this work, qubits (qumodes) are shown as single (triple) wires.

This initial linear growth is strictly bounded by

$$0 \leq \text{Erf}(c/2) \int_{-\infty}^{\infty} dx e^{-x^2} |\sin(cx)| \leq \frac{2}{\sqrt{\pi}}, \quad (31)$$

where the lower and upper bounds are approached for $c = 0$ and $c \rightarrow \infty$, respectively.

Large time evolution ($\theta \gg 1$): In the large- θ limit, we obtain a square-root growth

$$e^{\mathcal{M}(W_0)} \simeq \sqrt{\theta} \mathcal{I}(c), \quad (32)$$

where $\mathcal{I}(c)$ is a non-trivial function of c . However, for $c \rightarrow \infty$, it is possible to show that

$$e^{\mathcal{M}(W_0)} \lesssim \left(\frac{128\sqrt{\pi}}{\Gamma(1/4)^4} \right) \sqrt{\theta}. \quad (33)$$

The scaling behavior

$$\mathcal{M} = \frac{1}{2} \ln(\theta) + \text{const}, \quad (34)$$

is also the upper limit (unitarity limit) attainable by single-mode operations of a single phase-space coordinate. See Appendix A for more details.

In Fig. 4, we show the mana (WLN) as a function of θ for several values of c , together with the small- and large- θ bounds in the limit $c \rightarrow \infty$. For the smaller value $c = 0.5$, the generated negativity remains negligible until relatively large evolution times, where quartic and higher-order terms in the cosine expansion become important. By contrast, for $c = 1$ and even more so for larger values of c , the obtained mana grows more rapidly and eventually approaches the asymptotic bound.

V. SIMULATION WITH PHYSICAL QUMODES

In the previous section, we considered ideal one- and two-qumode cosine gates. Here, we decompose these gates into elementary CV-DV operations that can be implemented on trapped-ion platforms.

A. Trotterization and postselection

Using Trotterization, the trigonometric gates can be approximated in terms of elementary CV-DV operations [63, 64]. Within its regime of validity, the resulting circuit reproduces the action of the exact cosine gate. The circuit that approximates the cosine gate up to corrections of order $\mathcal{O}(\theta^2)$ is shown in Fig. 5 using a single ancillary gate qubit. Here, we follow the convention of Ref. [64], which is unitarily equivalent to the construction of Ref. [63]. See Appendix C for details. The circuit in Fig. 5 corresponds to the following CV-DV hybrid unitary

$$\begin{aligned} \hat{\mathcal{C}}_1(\theta, c) \equiv & \text{CD}_x \left(i \frac{c}{2\sqrt{2}} \right) \cdot R_z(\theta) \cdot \text{CD}_x^\dagger \left(i \frac{c}{\sqrt{2}} \right) \\ & \cdot R_z(\theta) \cdot \text{CD}_x \left(i \frac{c}{2\sqrt{2}} \right), \end{aligned} \quad (35)$$

where the subscript denotes a first-order Trotter decomposition. Separating the qubit and qumode degrees of freedom, we can write

$$\hat{\mathcal{C}}_1(\theta, c) = \hat{G}_1(\theta, c) \otimes \mathbb{I} + \hat{G}_2(\theta, c) \otimes Z + \hat{B}(\theta, c) \otimes X, \quad (36)$$

with

$$\begin{aligned} \hat{G}_1(\theta, c) &= \cos^2 \left(\frac{\theta}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \cos(2c\hat{x}) \\ &= 1 + \mathcal{O}(\theta^2), \\ \hat{G}_2(\theta, c) &= -i \sin(\theta) \cos(c\hat{x}) \\ &= -i\theta \cos(c\hat{x}) + \mathcal{O}(\theta^2), \\ \hat{B}(\theta, c) &= -i \sin^2 \left(\frac{\theta}{2} \right) \sin(2c\hat{x}) = \mathcal{O}(\theta^2). \end{aligned} \quad (37)$$

The operators $\hat{G}_{1,2}$ contain the desired leading-order contribution, while $\hat{B}$ represents the unwanted component that first appears at order $\mathcal{O}(\theta^2)$. Therefore, in the following, we will refer to $\hat{G}_{1,2}$ and $\hat{B}$ as the “good” and “bad” components of the circuit, respectively. The approximation can be systematically improved by increasing the number of Trotter

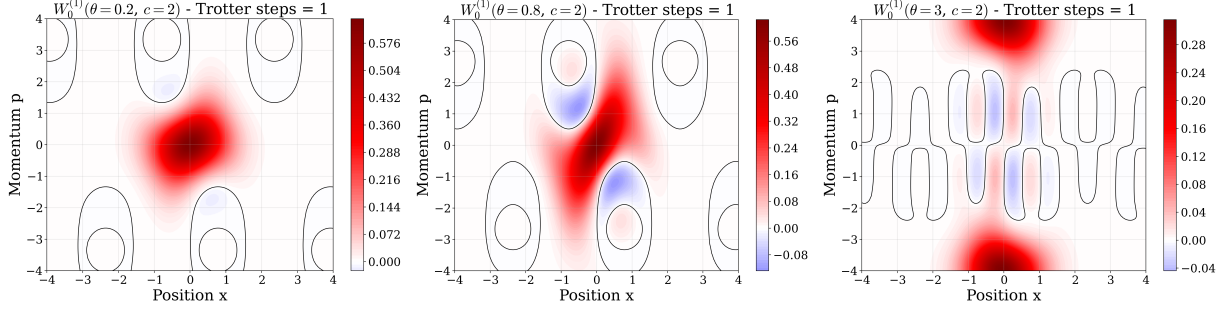


FIG. 6: Wigner function of the Fock vacuum $|0\rangle$ evolved for one Trotter step using the circuit in Fig. 5, after tracing out the qubit. The results are shown for different values of θ at fixed $c = 2$.

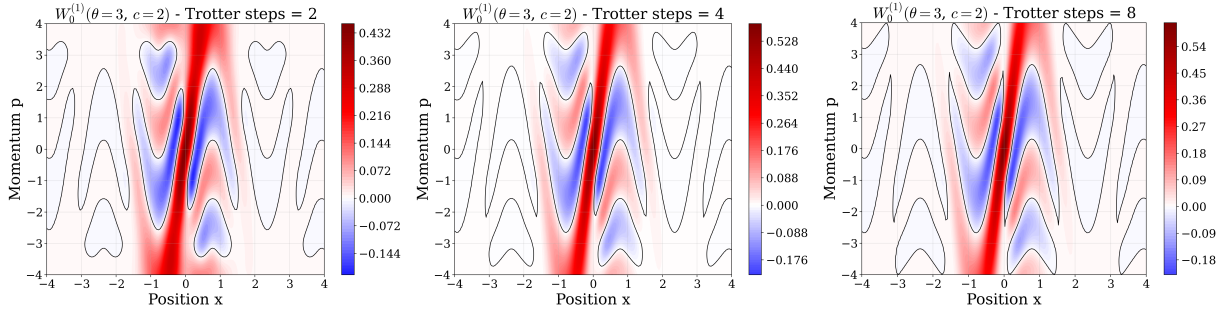


FIG. 7: Wigner function of the Fock vacuum $|0\rangle$ evolved for $n \in 2, 4, 8$ Trotter steps using the circuit in Fig. 5, after tracing out the qubit. The results are shown for fixed $\theta = 3$ and $c = 2$.

steps [82, 83]. For example, for two Trotter steps, we obtain

$$\begin{aligned} \hat{C}_2(\theta, c) &\equiv \hat{C}_1^2(\theta/2, c) \\ &= (\hat{G}_1^2 + \hat{G}_2^2 + \hat{B}^2) \otimes \mathbb{I} + 2\hat{G}_1\hat{G}_2 \otimes Z + 2\hat{G}_1\hat{B} \otimes X, \end{aligned} \quad (38)$$

where the argument θ of each bosonic operator is replaced by $\theta \mapsto \theta/2$. After a large number of Trotter steps, the circuit will asymptote to

$$\begin{aligned} \hat{C}_\infty(\theta, c) &\equiv \lim_{N \rightarrow \infty} \prod_{n=1}^N \hat{C}_1(\theta/N, c) \\ &= \cos(\theta \cos(c\hat{x})) \otimes \mathbb{I} - i \sin(\theta \cos(c\hat{x})) \otimes Z + 0 \otimes X \\ &= e^{-i\theta \cos(c\hat{x}) \otimes Z}. \end{aligned} \quad (39)$$

The number of conditional displacement gates CD_x after n Trotter steps is $\#\text{CD}_x = 2n+1$. More details about the systematic improvement can be found in Appendix B. The circuit performance can be improved by postselecting on the measurement outcome of the gate qubit: if a qubit flip is observed, the corresponding run is discarded.

If we focus on the single first-order Trotter step and initialize the qubit to the $|\uparrow\rangle$ state, the resulting

state can be written as

$$\hat{C}_1(\theta, c) |\psi\rangle |\uparrow\rangle = \hat{G} |\psi\rangle |\uparrow\rangle + \hat{B} |\psi\rangle |\downarrow\rangle, \quad (40)$$

where $|\psi\rangle$ is the initial state of the qumode and $\hat{G} \equiv \hat{G}_1 + \hat{G}_2$. Looking back at the results in Eq. (37), we can write the good and bad components in terms of displacement operators

$$\begin{aligned} \hat{G}_1(\theta, c) &= \cos^2\left(\frac{\theta}{2}\right) \mathbb{I} - \frac{1}{2} \sin^2\left(\frac{\theta}{2}\right) \left[D(i\sqrt{2}c) + D^\dagger(i\sqrt{2}c) \right], \\ \hat{G}_2(\theta, c) &= -\frac{i}{2} \sin(\theta) \left[D\left(i\frac{c}{\sqrt{2}}\right) + D^\dagger\left(i\frac{c}{\sqrt{2}}\right) \right], \\ \hat{B}(\theta, c) &= -\frac{1}{2} \sin^2\left(\frac{\theta}{2}\right) \left[D(i\sqrt{2}c) - D^\dagger(i\sqrt{2}c) \right]. \end{aligned} \quad (41)$$

The density matrix of the oscillator state $|\psi\rangle$ and qubit q system after one Trotter step is

$$\begin{aligned} \hat{\rho}_{\psi,q}^{(1)}(\theta, c) &\equiv \hat{C}_1(\theta, c) |\psi\rangle |\uparrow\rangle \langle\uparrow| \langle\psi| \hat{C}_1^\dagger(t, c) \\ &= \begin{pmatrix} \hat{G} |\psi\rangle \langle\psi| \hat{G}^\dagger & \hat{G} |\psi\rangle \langle\psi| \hat{B}^\dagger \\ \hat{B} |\psi\rangle \langle\psi| \hat{G}^\dagger & \hat{B} |\psi\rangle \langle\psi| \hat{B}^\dagger \end{pmatrix}. \end{aligned} \quad (42)$$

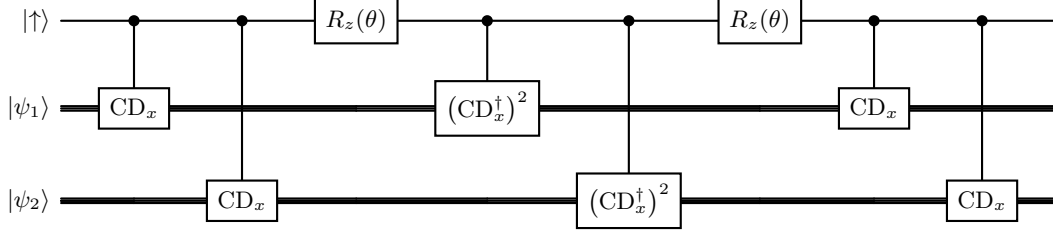


FIG. 8: Circuit that approximates the action of the trigonometric operator $e^{-i\theta \cos(c_1 \hat{x}_1 + c_2 \hat{x}_2)}$ acting on the two-qumode state $|\psi_1\rangle \otimes |\psi_2\rangle$ up to corrections of order $\mathcal{O}(\theta^2)$. The argument of the sequences conditional displacement is $\text{CD}_x = \text{CD}_x(ic_j/(2\sqrt{2}))$, with $c_j = c_1, c_2$.

Tracing out the qubit gives the mixed qumode state described by the following density matrix

$$\hat{\rho}_\psi^{(1)}(\theta, c) \equiv \text{Tr}_q [\hat{\rho}_{\psi,q}^{(1)}] = \hat{G} |\psi\rangle \langle \psi| \hat{G}^\dagger + \hat{B} |\psi\rangle \langle \psi| \hat{B}^\dagger. \quad (43)$$

The Wigner function of this reduced state can be computed as

$$W_\psi^{(1)}(\alpha; \theta, c) \equiv \frac{2}{\pi} \text{Tr}[\hat{\rho}_\psi^{(1)}(\theta, c) \text{D}(\alpha) \hat{\Pi} \text{D}^\dagger(\alpha)], \quad (44)$$

with $\alpha = (x + ip)/\sqrt{2}$ and $\hat{\Pi} \equiv (-1)^{\hat{a}^\dagger \hat{a}}$ is the parity operator.

In Fig. 6, we illustrate the Wigner function for the case where the state is initialized in the Fock vacuum, $|\psi\rangle = |0\rangle$. The dashed curves in the left panel of Fig. 4 are obtained by integrating the absolute value of Eq. (44), and represent the mana of the first-order Trotter approximation. In Fig. 7, we show the Wigner function of the evolved vacuum for $n \in \{2, 4, 8\}$ Trotter steps at fixed $\theta = 3$ and $c = 2$. As expected, increasing the number of Trotter steps drives the state toward the exact result shown in Fig. 2c.

B. Transition probabilities

If we initialize the mode to the zero Fock state $|\psi\rangle = |n=0\rangle$, the transition probabilities to other Fock states are

$$P_{0 \rightarrow n} = \langle n | \hat{\rho}_0^{(1)}(\theta, c) | n \rangle = |\langle n | \hat{G} | 0 \rangle|^2 + |\langle n | \hat{B} | 0 \rangle|^2 \\ = \begin{cases} |f_0(\theta) \delta_{n,0} - f_1(\theta, n)|^2 & n \text{ even,} \\ \sin^4\left(\frac{\theta}{2}\right) \frac{(2c^2)^n}{n!} e^{-2c^2} & n \text{ odd,} \end{cases} \quad (45)$$

with $f_0(\theta) = \cos^2(\frac{\theta}{2})$ and

$$f_1(\theta, n) = \frac{(ic)^n}{\sqrt{n!}} \left(\sin^2\left(\frac{\theta}{2}\right) \frac{e^{-c^2}}{2^{-\frac{n}{2}}} + i \sin(\theta) \frac{e^{-\frac{c^2}{4}}}{2^{\frac{n}{2}}} \right).$$

The structure of Eq. (45) reflects a useful parity property of the first-order circuit: $\hat{G}$ connects the vacuum only to even Fock states, while $\hat{B}$ connects it only to odd Fock states. As a result, postselecting on the qubit state $|\uparrow\rangle$ leaves the even transition probabilities unchanged, while removing the odd transitions. The same parity-based interpretation extends to multiple Trotter steps, although the explicit form of the transition probabilities becomes more involved.

Equivalently, the bosonic component supported on the $|\uparrow\rangle$ qubit state contains the parity-preserving contribution associated with the exact trigonometric gate, whereas the component supported on the $|\downarrow\rangle$ qubit state contains the parity-violating contribution generated by the finite-step approximation.

We now extend the construction to trigonometric gates whose argument is a weighted sum of commuting Hermitian operators. We focus on the minimal two-mode case, which is the relevant setting for the experiments presented below. Specifically, we consider the gate

$$e^{-i\theta \cos(c_1 \hat{x}_1 + c_2 \hat{x}_2)} = e^{-i\theta \cos(\mathbf{c} \cdot \hat{\mathbf{x}})}, \quad (46)$$

with $\mathbf{c} = (c_1, c_2) \in \mathbb{R}^2$ and $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2)$ denotes two independent position quadratures. From Refs. [63, 64], we obtain the circuit shown in Fig. 8. The extension to additional commuting quadratures is obtained by applying the corresponding conditional displacements to each mode using the same control qubit.

The circuit in Fig. 8, which approximates the cosine gate of a sum of two quadratures, can be analyzed in terms of the operators that compose it. In

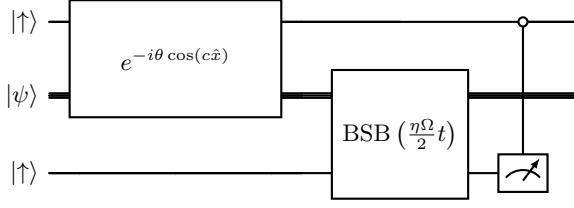


FIG. 9: Qumode-readout circuit. After applying the hybrid trigonometric gate acting on the qumode and gate qubit, the qumode is coupled to the probe qubit through a blue sideband gate, see Eq. (5), and the probe qubit's probability $P_{\downarrow}(t)$ is measured when the gate qubit remains in $|\uparrow\rangle$.

analogy with Eq. (35), we define the hybrid unitary

$$\begin{aligned} \hat{C}_{1,(2)}(\theta, \mathbf{c}) \equiv & \prod_{j=1}^2 \text{CD}_{x,j} \left(i \frac{c_j}{2\sqrt{2}} \right) \cdot R_z(\theta) \cdot \prod_{j=1}^2 \text{CD}_{x,j}^\dagger \left(i \frac{c_j}{\sqrt{2}} \right) \\ & \cdot R_z(\theta) \cdot \prod_{j=1}^2 \text{CD}_{x,j} \left(i \frac{c_j}{2\sqrt{2}} \right). \end{aligned} \quad (47)$$

This unitary can be decomposed as

$$\begin{aligned} \hat{C}_{1,(2)}(\theta, \mathbf{c}) = & \hat{G}_{1,(2)}(\theta, \mathbf{c}) \otimes \mathbb{I} + \hat{G}_{2,(2)}(\theta, \mathbf{c}) \otimes Z + \hat{B}_{(2)}(\theta, \mathbf{c}) \otimes X, \end{aligned} \quad (48)$$

with

$$\begin{aligned} \hat{G}_{1,(2)}(\theta, \mathbf{c}) &= \cos^2 \left(\frac{\theta}{2} \right) - \sin^2 \left(\frac{\theta}{2} \right) \cos(2\mathbf{c} \cdot \hat{\mathbf{x}}) = 1 + \mathcal{O}(\theta^2), \\ \hat{G}_{2,(2)}(\theta, \mathbf{c}) &= -i \sin(\theta) \cos(\mathbf{c} \cdot \hat{\mathbf{x}}) = -i\theta \cos(\mathbf{c} \cdot \hat{\mathbf{x}}) + \mathcal{O}(\theta^2), \\ \hat{B}_{(2)}(\theta, \mathbf{c}) &= -i \sin^2 \left(\frac{\theta}{2} \right) \sin(2\mathbf{c} \cdot \hat{\mathbf{x}}) = \mathcal{O}(\theta^2). \end{aligned} \quad (49)$$

The interpretation is analogous to the one-mode case. We emphasize that, for transitions out of the two-mode vacuum $|0\rangle_1 \otimes |0\rangle_2$, the ideal gate conserves the total parity rather than the parity of each individual mode. For example, the transition $|0\rangle_1 \otimes |0\rangle_2 \rightarrow |1\rangle_1 \otimes |1\rangle_2$ is allowed because the total parity is preserved, unlike in the single-mode case where the transition from the vacuum to the first Fock state is forbidden by parity. Similarly to the one-mode case, the total-parity-violating

bosonic component $\hat{B}_{(2)}|0\rangle_1|0\rangle_2$ has support only on the $|\downarrow\rangle$ qubit state. Therefore, postselecting on the $|\uparrow\rangle$ qubit state removes contributions such as $|0\rangle_1 \otimes |0\rangle_2 \rightarrow |1\rangle_1 \otimes |0\rangle_2$, which violate total parity conservation.

VI. EXPERIMENTAL RESULTS WITH TRAPPED IONS

In this section, we discuss the qumode readout and the modeling of trapped ion device errors. We then present experimental results for the trigonometric gates introduced above.

A. Qumode measurement

As discussed in section II, we do not directly measure the state of the bosonic modes. Instead, we reconstruct the Fock-state populations using the ancillary probe qubit coupled to the qumode under consideration. Denoting the qumode state by $|\psi(\theta)\rangle$, we apply the BSB gate from Eq. (5) to the combined qubit-qumode state,

$$e^{it\frac{\eta\Omega}{2}(a\sigma^- + a^\dagger\sigma^+)} |\uparrow, \psi(\theta)\rangle, \quad (50)$$

where η is the Lamb-Dicke parameter, and Ω is the carrier Rabi frequency. The probability of measuring the probe qubit in the state $|\downarrow\rangle$ is then

$$P_{\downarrow}(t, \theta) = \sum_{n=0}^{\Lambda} |\langle \downarrow, n | e^{it\frac{\eta\Omega}{2}(a\sigma^- + a^\dagger\sigma^+)} |\uparrow, \psi(\theta)\rangle|^2. \quad (51)$$

Here, the cutoff Λ specifies the maximal Fock occupation retained in the classical simulation and reflects the finite-dimensional experimental reconstruction, where only a limited range of occupation numbers can be resolved reliably. Instead of treating the cutoff as a fit parameter, we fix $\Lambda = 40$ throughout this work unless specified otherwise, which we find provides a good description of the data. For ideal BSB evolution, this signal consists of a sum of Rabi oscillations with frequencies proportional to $\sqrt{n+1}$, whose amplitudes are the Fock-state populations $P_n(\theta) = |\langle n | \psi(\theta) \rangle|^2$. See Fig. 9 for an illustration of the qumode readout circuit. To account for dephasing during the BSB readout, we replace the unitary evolution in Eq. (53) by Lindblad evolution with dephasing rate 4Γ (more details about

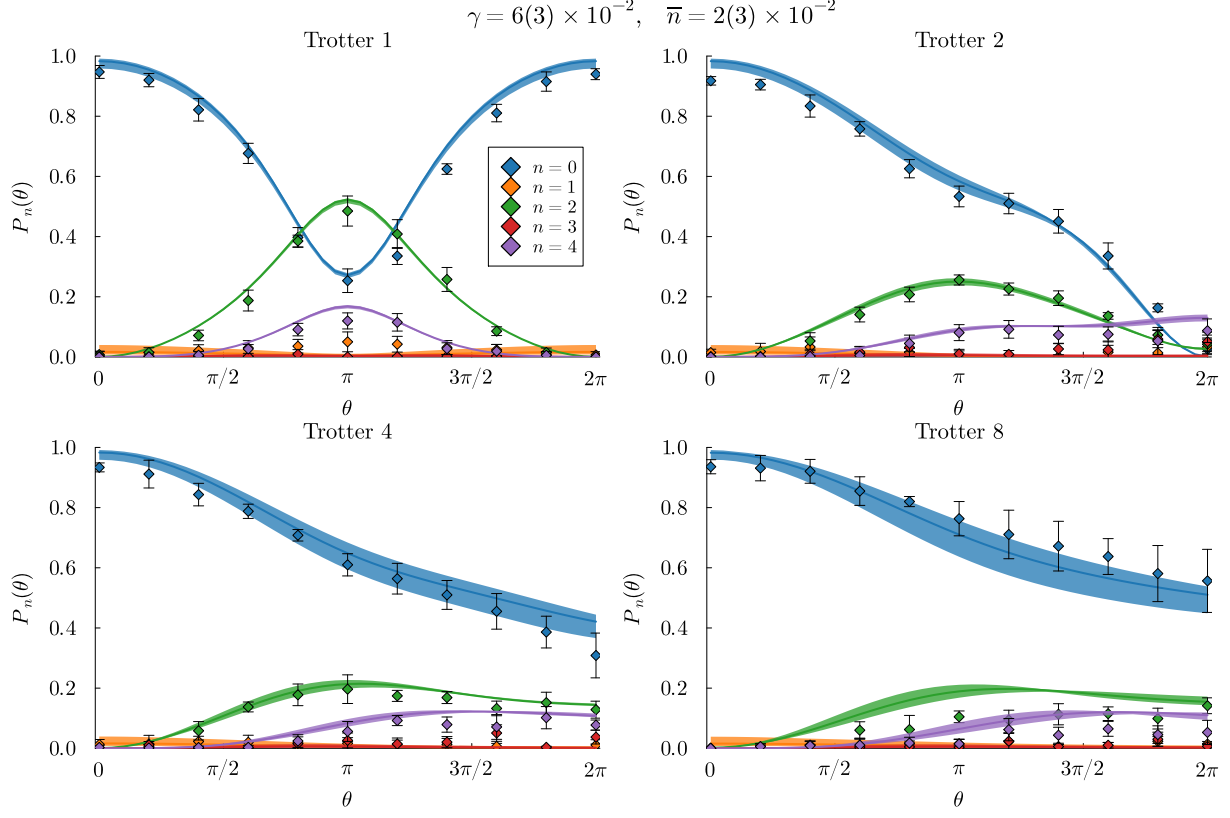


FIG. 10: Postselected transition probabilities as a function of θ for the $c = 1$ circuit Fig. 5 (and its higher Trotter step variants) using Trotter steps of 1 (top left), 2 (top right), 4 (bottom left), and 8 (bottom right). The data (diamonds) is plotted alongside the optimized simulations (lines with uncertainty bands). The parameters that optimize the simulation are displayed at the top of each plot.

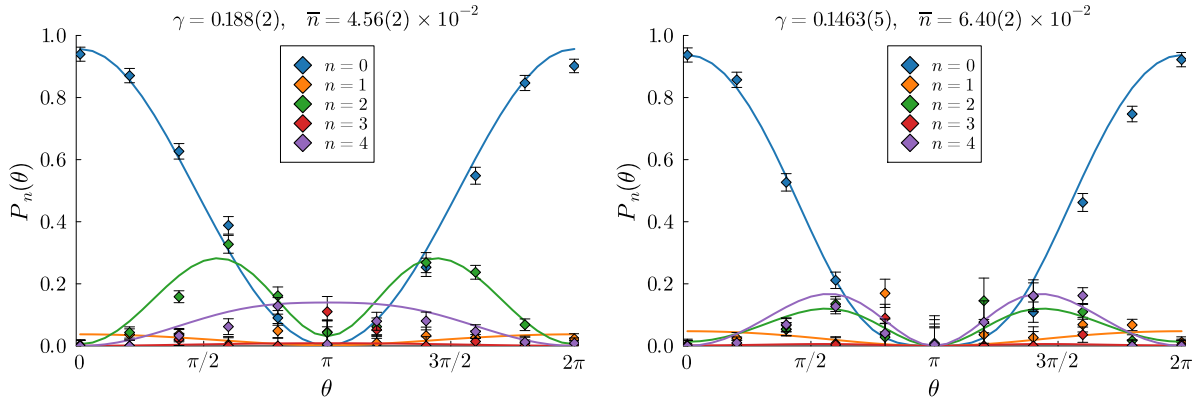


FIG. 11: Postselected transition probabilities as a function of θ for the $c = 2$ (left) and $c = 3$ (right), Trotter one circuit Fig. 5. The width of the uncertainties of the simulations (lines) are too small to resolve.

Lindbladian dynamics are addressed in the next sub-

section (VIB). With this convention, we find

$$P_{\downarrow}(t, \theta) = \frac{1}{2} \left[1 - \sum_{n=0}^{\Lambda} P_n(\theta) e^{-\Gamma t} \left(\cos(\tilde{\Omega}_n t) + \frac{\Gamma}{\tilde{\Omega}_n} \sin(\tilde{\Omega}_n t) \right) \right], \quad (52)$$

where $\tilde{\Omega}_n = \sqrt{\eta^2 \Omega^2 (n+1) - \Gamma^2}$. In the regime relevant for the experiment, $\Gamma \ll \eta\Omega$, the readout signal is well approximated by

$$P_{\downarrow}(\theta, t) = \frac{1}{2} \left[1 - \sum_{n=0}^{\Lambda} P_n(\theta) e^{-\Gamma t} \cos(t\eta\Omega\sqrt{n+1}) \right] + \mathcal{O}\left(e^{-\Gamma t} \frac{\Gamma}{\eta\Omega}\right). \quad (53)$$

For each value of $\theta = (2\pi/10)m$ with integer $m \in [0, 10]$, the readout qubit in Eq. (53) is measured repeatedly for each BSB evolution time t . The BSB time is varied from 0 to 300 μs for the single qumode case using 101 uniformly spaced values. For the two qumode circuit, the modes have different frequencies and are sampled for a total of 8π oscillations with 101 uniformly spaced values. In addition, the circuit data is postselected on the gate qubit being measured in $|\uparrow\rangle$. The Fock-state probabilities $P_n(\theta)$ are reconstructed by fitting Eq. (53) for $n \in [0, 20]$, with the readout dephasing rate Γ and the effective sideband coupling $\eta\Omega$ included as fit parameters. The fit minimizes the loss function,

$$\begin{aligned} \mathcal{L}(\{P_n(\theta)\}, \eta\Omega, \Gamma) &= \frac{\sum_{\ell} w(\theta, t_{\ell}) [P_{\downarrow, \text{data}}(\theta, t_{\ell}) - P_{\downarrow}(\theta, t_{\ell}; \{P_n(\theta)\}, \eta\Omega, \Gamma)]^2}{\sum_{\ell} w(\theta, t_{\ell})}, \\ (\{P_n(\theta)\}, \eta\Omega, \Gamma)_{\text{opt}} &= \arg \min_{\{P_n(\theta)\}, \eta\Omega, \Gamma} \mathcal{L}(\{P_n(\theta)\}, \eta\Omega, \Gamma). \end{aligned} \quad (54)$$

where ℓ indexes each measured BSB time, $P_{\downarrow}$ is given in Eq. (53), $P_{\downarrow, \text{data}}$ is the mean of the experimentally measured readout qubit spin, $w(\theta, t_{\ell}) = 1/\sigma(\theta, t_{\ell})^2$ is the square reciprocal of the statistical binomial variance arising from the sampling of the readout qubit,

$$\sigma(\theta, t_{\ell}) = \sqrt{P_{\downarrow, \text{data}}(\theta, t_{\ell})(1 - P_{\downarrow, \text{data}}(\theta, t_{\ell}))/N_{\uparrow}}, \quad (55)$$

where $N_{\uparrow}$ is the number of experimental shots that survive postselection. Typical fitted values of Γ and $\eta\Omega$ are in the ranges $2\pi \times (0.095 - 0.0014) \mu\text{s}^{-1}$ and $2\pi \times (0.0167 - 0.0175) \mu\text{s}^{-1}$, respectively, justifying the approximation in Eq. (53). For each circuit and each value of θ , the data are fit as a function of the BSB evolution time, from which the transition probabilities $P_n(\theta)$ are reconstructed. The quoted uncertainties on the reconstructed probabilities are the standard errors obtained from the fit. For each value of θ and each BSB evolution time, we collected

1200 shots for the $c = 1$ circuits with $i = 1, 2, 4$ Trotter steps, and 1000 shots for the corresponding $i = 8$ -Trotter-step circuit. For the one-Trotter-step circuits with $c = 2, 3$, as well as for each two-qumode circuit considered, we collected 200 shots per value of θ and per BSB evolution time.

All classical calculations reported in this work are performed using `QuantumToolbox.jl` [84] in Julia [85]. For computational efficiency, we use a reduced cutoff of $\Lambda = 15$ in the two-qumode simulations.

B. Modeling device errors

We now describe the error model used to compare the experimental data with classical simulations. We distinguish between initialization errors, which affect the prepared motional state before the circuit is applied, dynamical errors, which arise during the implementation of the qubit-qumode gates and calibration errors, which are caused by a erroneous calibration of the conditional displacement's parameters. Readout errors were discussed separately in the previous section.

Initialization errors: In an ideal implementation of the circuit in Eq. (40), as well as of the subsequent Trotter steps, the qumode would be initialized in the pure vacuum state $|0\rangle$. In practice, however, the motional mode can have a small residual thermal occupation. We therefore model the initial qumode state as a thermal state,

$$\begin{aligned} \hat{\rho}(\bar{n}) &= \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(1 + \bar{n})^{1+n}} |n\rangle\langle n| \\ &= (1 - e^{-\beta\omega}) \sum_{n=0}^{\infty} e^{-n\beta\omega} |n\rangle\langle n|, \end{aligned} \quad (56)$$

where $\bar{n}$ denotes the mean occupation number, or average phonon number, of the qumode. It is related to the inverse temperature β and energy ω (in natural units) of the qumode as $\bar{n} = (e^{\beta\omega} - 1)^{-1}$. Note that $\bar{n}$ also quantifies the purity of the state, which is given by

$$\text{Tr}(\hat{\rho}(\bar{n})^2) = \tanh\left(\frac{\beta\omega}{2}\right) = \frac{1}{2\bar{n} + 1} \leq 1. \quad (57)$$

Dynamical errors: Circuit errors are dominated by qumode dephasing. Motional heating rates for the addressed radial modes are approximately 10 quanta/s on the zig-zag mode and 100 quanta/s

on the drum mode, contributing negligible additional occupation over the duration of a typical experimental sequence. The dominant error source is instead motional dephasing, for which coherence times in the trap are 500–700 μs . Because qubit rotations are implemented on timescales much shorter than the qumode gates, we model circuit errors at the level of the hybrid qubit-qumode gates, replacing the ideal unitary evolution of each conditional displacement with open quantum system Lindblad dynamics.

The Lindblad master equation is given by

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \sum_i \left(\hat{C}_i \hat{\rho} \hat{C}_i^\dagger - \frac{1}{2} \{ \hat{C}_i^\dagger \hat{C}_i, \hat{\rho} \} \right), \quad (58)$$

where $\hat{C}_i = \sqrt{\gamma_i} \hat{L}_i$ are collapse operators, and γ_i sets the strength of the corresponding dissipative process. We take $\hat{L} = \hat{n} = \hat{a}^\dagger \hat{a}$, which corresponds to qumode dephasing. Physically, this noise channel can be interpreted as a continuous weak measurement of the qumode occupation number by the environment. Experimentally, this dephasing likely is the result of fast fluctuations in the motional mode frequency as the result of noise on the RF and DC grounds, leading to a stochastic sampling of gate detunings from the ideal. A Fock state $|n\rangle$ acquires phase at a rate proportional to the instantaneous detuning, integrating over these fluctuations results in phase diffusion in the Fock basis described by $\hat{L}$.

For the conditional displacement gate CD_x , the gate Hamiltonian is $\hat{H}_+ = -\hat{x} \otimes \hat{X}$, integrated for duration $t = c/2$. For the gate $(\text{CD}_x^\dagger)^2$ the Hamiltonian sign is reversed, $\hat{H}_- = +\hat{x} \otimes \hat{X}$, and evolves for twice the duration, $t = c$. This choice avoids negative evolution times while producing the conjugate displacement.

The initial state of the qubit-qumode system is

$$\hat{\rho}_{\text{in}} \equiv \hat{\rho}(\bar{n}) \otimes |\uparrow\rangle\langle\uparrow|, \quad (59)$$

where $\hat{\rho}(\bar{n})$ is the thermal density matrix with mean occupation $\bar{n}$, see Eq. (56). The Lindblad-deformed evolution of the i -th Trotter circuit is represented by the quantum channel $\mathcal{G}_{\hat{C}_i}^{(\gamma)}$, the open quantum system counterpart of the ideal unitary $\hat{C}_i^i(\theta, c)$: in the absence of dephasing ($\gamma \rightarrow 0$) the channel reduces to conjugation by $\hat{C}_i^i$. The state after the noisy circuit is therefore

$$\hat{\rho}_{\mathcal{G}}^{(i)} \equiv \mathcal{G}_{\hat{C}_i}^{(\gamma)}[\hat{\rho}_{\text{in}}], \quad (60)$$

and the normalized Fock-state transition probabilities are

$$P_n^{(i)}(\theta, c; \gamma, \bar{n} | \uparrow) = \frac{\langle n, \uparrow | \hat{\rho}_{\mathcal{G}}^{(i)} | n, \uparrow \rangle}{\sum_{n'=0}^{\Lambda} \langle n', \uparrow | \hat{\rho}_{\mathcal{G}}^{(i)} | n', \uparrow \rangle}. \quad (61)$$

The projection onto $|\uparrow\rangle$ mirrors the experimental postselection procedure, and the denominator normalizes the postselected distribution to unity.

For the two-qumode simulation, the initial mean phonon occupations $\bar{n}_1, \bar{n}_2$ and dephasing rates γ_1, γ_2 are treated as independent parameters. The sequential application of CD_x gates to the two modes (Fig. 8) breaks the symmetry between the marginal transition probabilities of the two qumodes when dephasing is present. This asymmetry vanishes in the noiseless limit $\gamma_i = 0$.

With two qumodes, the initial state is

$$\hat{\rho}_{\text{in}} = \hat{\rho}(\bar{n}_1) \otimes \hat{\rho}(\bar{n}_2) \otimes |\uparrow\rangle\langle\uparrow|, \quad (62)$$

and the state after the i -th noisy Trotter step is

$$\hat{\rho}_{\mathcal{G}}^{(i)} \equiv \mathcal{G}_{\hat{C}_i}^{(\gamma_1, \gamma_2)}[\hat{\rho}_{\text{in}}], \quad (63)$$

where $\mathcal{G}_{\hat{C}_i}^{(\gamma_1, \gamma_2)}$ is the two-mode quantum channel with independent per-mode dephasing rates, and reduces to conjugation by $\hat{C}_i^i$ when $\gamma_1 = \gamma_2 = 0$. Postselecting on the qubit in $|\uparrow\rangle$, the marginal transition probability for qumode 1 is obtained by tracing out qumode 2

$$P_{n_1}^{(i)}(\theta, c; \gamma_1, \gamma_2, \bar{n}_1, \bar{n}_2 | \uparrow) = \frac{\text{Tr} \left[(|n_1\rangle\langle n_1|_1 \otimes \mathbb{I}_2 \otimes |\uparrow\rangle\langle\uparrow|) \hat{\rho}_{\mathcal{G}}^{(i)} \right]}{\text{Tr} \left[(\mathbb{I}_{12} \otimes |\uparrow\rangle\langle\uparrow|) \hat{\rho}_{\mathcal{G}}^{(i)} \right]}. \quad (64)$$

Here $\mathbb{I}_{12}$ is the identity on the joint qumode Hilbert space, and the denominator is the postselection probability normalizing the marginal distribution to unity. The marginal transition probabilities for qumode 2 follow analogously from Eq. (64) with the roles of the two modes exchanged.

With these assumptions, the classical simulations of the Lindblad-deformed circuit depend only on a small set of parameters: the initial mean phonon occupations $\{\bar{n}_1, \bar{n}_2\}$ and the dephasing rates $\{\gamma_1, \gamma_2\}$. The optimal parameters are obtained by minimizing a weighted least-squares loss between the simulated probabilities and those extracted from the experimental data. Note that this is a second fitting stage,

$$\gamma_1 = 1.0(3), \gamma_2 = 1.1(6), \bar{n}_1 = 0.14(3), \bar{n}_2 = 9(5) \times 10^{-2}$$

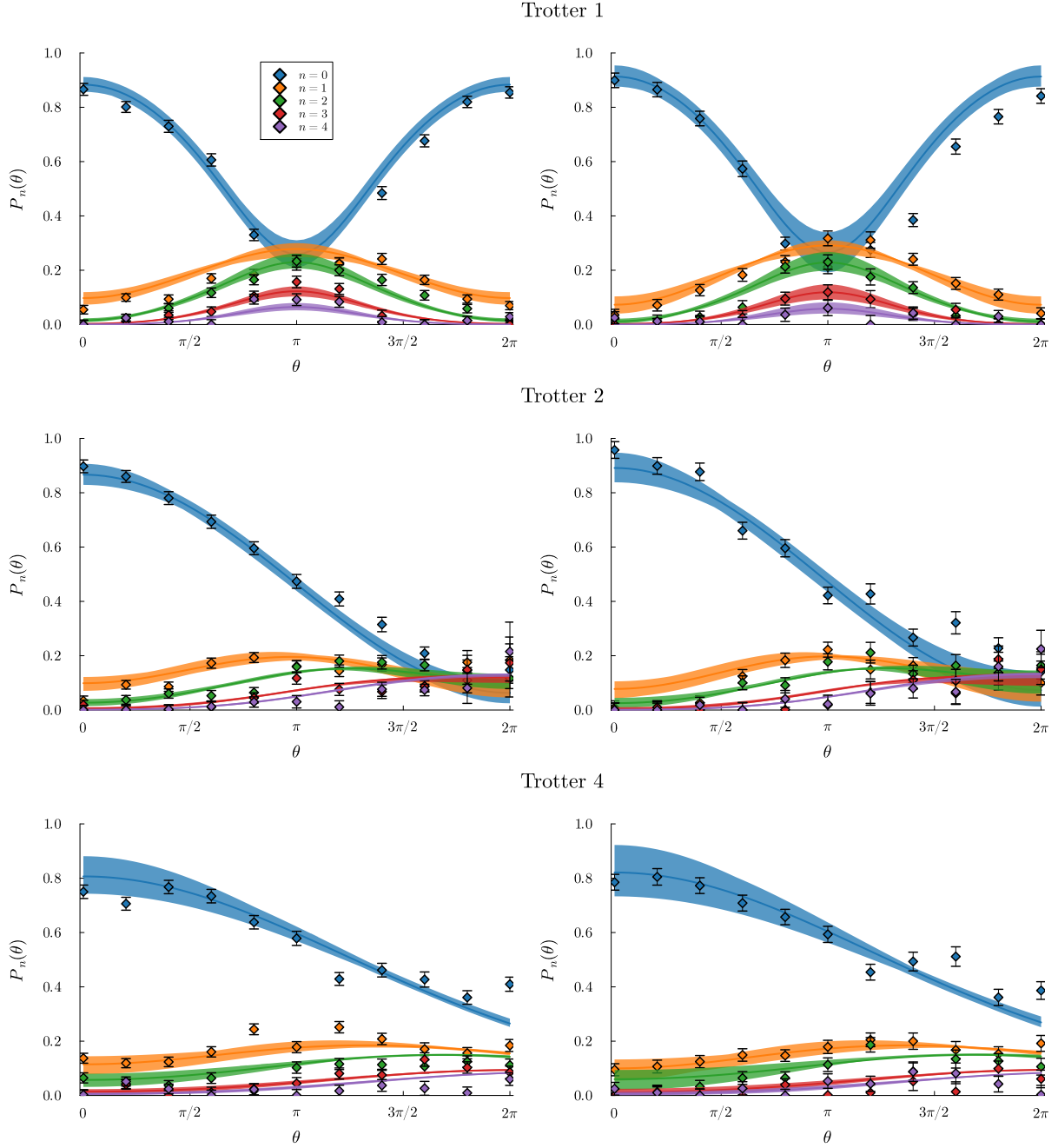


FIG. 12: Data for the transition probabilities $P_n(\theta)$ of the circuit in Fig. 8 (and its higher Trotter order variants) for qumode 2 (left column) and qumode 3 (right column) on the QSCOUT platform, using one (top row), two (middle row) and four (bottom row) Trotter steps. The optimal dephasing and the initial phonon occupation of the corresponding mode are labeled atop the plots.

distinct from the BSB-readout fit used to extract the experimental probabilities as discussed in the previ-

ous section:

$$\mathcal{L}(\{\bar{n}\}, \{\gamma\}) = \frac{\sum_{\ell} w_{\ell} [P_{\text{data}, \ell} - P_{\text{sim}, \ell}(\{\bar{n}\}, \{\gamma\})]^2}{\sum_{\ell} w_{\ell}},$$

$$(\{\bar{n}\}, \{\gamma\})_{\text{opt}} = \arg \min_{\{\bar{n}\}, \{\gamma\}} \mathcal{L}(\{\bar{n}\}, \{\gamma\}). \quad (65)$$

Here ℓ labels the fitted data points, including the values of θ , the measured Fock states, and, when applicable, the qumode index. The weights are chosen as $w_\ell = 1/\sigma_\ell^2$, where σ_ℓ is the standard error of the corresponding probability extracted from experimental data. We verify convergence to a minimum by evaluating the Hessian of the loss function at the optimum and checking that its eigenvalues are positive within numerical precision.

Calibration errors: Another error source is the miscalibration of the motional mode frequency ($\delta \sim 0.1 - 1$ kHz) in the conditional displacement operator, governed by the Hamiltonian in Eq. (9). A nonvanishing detuning δ rotates the conditional displacement axis in phase space, replacing the target operation $\text{CD}_x(\alpha)$ with $\text{CD}_x(\alpha e^{i\varphi})$, where $\varphi = -\delta t$, the product of the detuning and the gate duration. The Fock transition probabilities are insensitive to φ . To see this, note that the action of the phase rotation $\alpha \mapsto \alpha e^{i\varphi}$ is equivalent to conjugating the k -th Trotter circuit (the k -th analogue of Eq. (38)) as

$$\hat{\mathcal{C}}_1^k(\varphi) \equiv \hat{R}^\dagger(\varphi) \hat{\mathcal{C}}_1^k \hat{R}(\varphi). \quad (66)$$

Since $\hat{R}(\varphi)$ is diagonal in the Fock basis, with $\hat{R}(\varphi)|n\rangle = e^{in\varphi}|n\rangle$, its effect on matrix elements is a state-dependent global phase

$$\langle m | \hat{\mathcal{C}}_1^k(\varphi) | n \rangle = e^{i(n-m)\varphi} \langle m | \hat{\mathcal{C}}_1^k | n \rangle. \quad (67)$$

This phase cancels upon taking the modulus squared, leaving all transition probabilities independent of φ . While the Fock transition probabilities are unaffected, the Wigner and characteristic functions of the qumode are sensitive to φ . Since $\hat{R}(\varphi)$ acts as a rigid rotation in phase space, the Wigner function transforms as $W(\alpha) \mapsto W(\alpha e^{i\varphi})$, and likewise for the characteristic function $\chi(\beta) \mapsto \chi(\beta e^{i\varphi})$. We take this effect into account when comparing the experimental measurement below to the theoretical calculation.

C. Transition matrix elements

Using the BSB readout of the qumodes, we reconstruct transition probabilities for the one- and two-qumode circuits at several Fock levels n and Trotter steps i , and compare them to classical simulations with fitted dephasing rates and initial thermal phonon occupations. The results for the one-qumode circuit with parameter $c = 1$ and Trotter

steps $i = 1, 2, 4, 8$ are shown in Fig. 10. Overall, we find good agreement between the experimental data and the simulations. The agreement is strongest for the transition to the lowest Fock level, corresponding to $\rho(\bar{n}) \rightarrow |0\rangle\langle 0|$, which has the strongest signal overall, while transitions to higher Fock levels can show somewhat larger deviations. The optimized values of the dephasing rate γ and thermal occupation $\bar{n}$ are obtained via a global fit to the four circuits, and are displayed at the top, with γ given in units of the reciprocal of the physical CD_x gate time. Each Trotter step comprises three CD_x gates, the second of which is twice the duration of the other two, so that a single step takes the equivalent of four CD_x gate times. The R_z gate in each step contributes negligibly to the circuit duration.

Data and simulations for the one-Trotter-step circuits with parameters $c = 2, 3$ are shown in Fig. 11. The optimized parameters are again consistent with the experimental conditions. In this case, the circuit structure is fixed while the gate strength is varied, such that the fitted dephasing rates are expected to be more directly comparable. The fitted phonon occupations are also consistent with the one-qumode results in Fig. 10. The fits are done for each individual circuit, and achieve much stronger agreement. This gives small uncertainties in the parameters and therefore (very) small uncertainty bands on the simulation curves.

The marginal Fock-state distributions for the two-qumode circuit with $c = 1$ and $i = 1, 2, 4$ Trotter steps are shown in Fig. 12, together with the corresponding optimized simulations. The simulations generally reproduce the measured distributions, including the nontrivial population transfer to higher Fock levels, indicating that the error model captures the dominant features of the trapped-ion implementation. For the two-qumode data, the BSB readout yields mode-resolved marginal Fock-state distributions rather than the full joint distribution. The measurement, therefore, verifies population transfer into each addressed mode, but does not directly measure correlations between the two modes. Nevertheless, the enhancement of odd Fock-level populations in the two-mode case, compared with their suppression in the one-mode results of Fig. 10, provides an indirect signature of the nontrivial two-mode dynamics shown in Fig. 12.

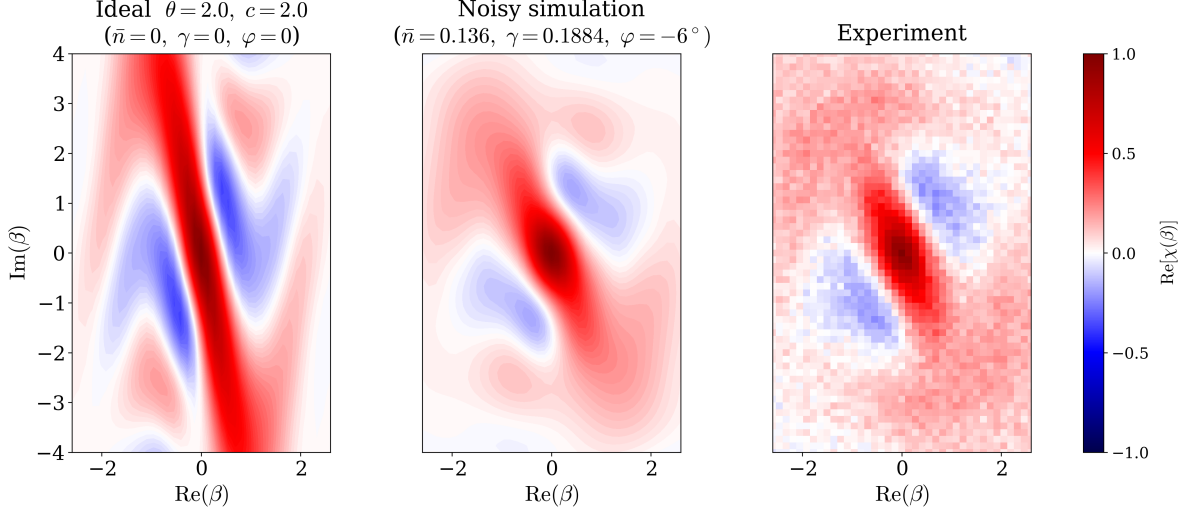


FIG. 13: Real part of the characteristic function $\chi(\beta) = \text{Tr}[\hat{\rho}_\psi \hat{D}(\beta)]$ of the post-selected qumode state obtained by applying the two-step Trotter approximation of the cosine gate, see Eq. (38), with $\theta = 2$, $c = 2$. Left: ideal simulation. Middle: simulation including qumode dephasing ($\gamma = 0.1884$), displacement miscalibration angle $\varphi = -6^\circ$ and thermal initialization ($\bar{n} = 0.136$). Right: Experimental reconstruction.

D. Characteristic function

Finally, in Fig. 13 we show the characteristic function $\chi(\beta) = \text{Tr}[\hat{\rho}_\psi \hat{D}(\beta)]$ (related to the Wigner function via Eq. (22)) of the qumode state obtained after post-selecting the gate qubit in $|\uparrow\rangle$, for two Trotter steps as in Eq. (38) at $\theta = 2$, $c = 2$. The left panel displays the ideal noiseless result ($\bar{n} = 0$, $\gamma = 0$, $\varphi = 0$), establishing the ideal target state structure. In the middle panel we consider the three dominant experimental imperfections:

- Qumode dephasing at rate $\gamma = 0.1884$, chosen as the one obtained from the fit in the left panel of Fig. 11.
- Miscalibration of the conditional displacements $\text{CD}_x(\alpha) \rightarrow \text{CD}_x(\alpha e^{i\varphi})$. For a gate duration of $t \approx 15 \mu\text{s}$ and an expected detuning of $\delta/2\pi \approx 750\text{--}1500 \text{ Hz}$, the accumulated phase error $-\varphi = \delta t$ is roughly $0.07\text{--}0.14 \text{ rad}$, which perfectly matches the $4^\circ\text{--}8^\circ$ expectation represented in Fig. 13 with the middle value $\varphi = -6^\circ$.
- A non-vanishing mean phonon occupation number measured as $\bar{n} = 0.136$ due to an imperfect initialization.

The resulting characteristic function is in good qualitative agreement with the experimentally recon-

structed one measured via the method described in [76] (right panel), confirming that qumode dephasing is the main source of deviation from the ideal case, followed by a small contribution due to the imperfect initialization and the miscalibration of the conditional displacement's argument.

VII. CONCLUSIONS AND OUTLOOK

We have experimentally demonstrated and benchmarked elementary trigonometric continuous-variable (CV) gate primitives on a hybrid qubit-qumode trapped-ion platform, providing the first gate-level characterization of these non-polynomial bosonic operations. The central objects studied are the one- and two-qumode cosine gates $e^{-i\theta \cos(c\hat{x})}$ and $e^{-i\theta \cos(c_1\hat{x}_1 + c_2\hat{x}_2)}$. These were implemented as finite-step Trotter circuits using conditional displacements and ancillary qubit rotations on the QS-COUT platform, with motional modes of $^{171}\text{Yb}^+$ ion chains serving as qumodes.

On the theory side, we derived analytical expressions for the ideal gate matrix elements, Wigner and characteristic functions, and Wigner logarithmic negativity, establishing how trigonometric gates generate non-Gaussian phase-space structure from simple initial states and placing them at the boundary of what single-mode non-Gaussian evolution can

produce from a Gaussian input. We also analyzed the finite-step Trotter approximations, identifying a decomposition into parity-preserving and parity-violating bosonic components that makes the role of postselection transparent: discarding runs in which the gate qubit is flipped removes all parity-violating contributions, a property that extends to total parity conservation in the two-mode case.

Experimentally, we reconstructed Fock-state transition probabilities via blue-sideband readout and compared them with open quantum system simulations incorporating residual thermal initialization and motional dephasing. For the one-qumode circuits, measurements across gate parameters $c \in \{1, 2, 3\}$ and Trotter step counts $i \in \{1, 2, 4, 8\}$ show good agreement with simulation, with distributions converging toward the exact gate with increasing Trotter depth. For the two-qumode circuit, the mode-resolved marginal distributions show nontrivial population transfer into both addressed modes, and the enhancement of odd-Fock-level populations, absent in the single-mode postselected data, provides a qualitative signature of genuinely two-mode dynamics. The reconstructed characteristic function of the one-qumode output state further confirms that motional dephasing is the dominant source of deviation from the ideal gate, with small additional contributions from residual thermal occupation and displacement miscalibration.

Looking ahead, improved motional coherence and displacement calibration would extend the accessible range of gate parameters and enable deeper Trotter sequences. For the two-qumode gate, direct characterization of intermode correlations would require simultaneous mode readout or heterodyne detection of motional modes, capabilities beyond current trapped-ion hardware that would provide a more complete picture of the two-mode dynamics. More broadly, the gates demonstrated here provide natural building blocks for hybrid quantum simulations of bosonic systems with periodic or compact structure, including molecular dynamics, rotor models, sine-Gordon dynamics, scalar field theories, and compact lattice gauge theories. In these settings, trigonometric gates offer a hardware-efficient, non-polynomial alternative to decompositions based on polynomial operations.

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Appendix A: Bound on the negativity growth

The most general one-mode pure Gaussian state is a squeezed displaced vacuum, and it can be fully parametrized by its position displacement x_0 , momentum displacement p_0 , spatial width parameter $a > 0$, and position-momentum correlation b . In the position representation, its wavefunction is given by

$$\psi_G(x) = \left(\frac{2a}{\pi}\right)^{1/4} e^{-(a+ib)(x-x_0)^2 + ip_0x}. \quad (\text{A1})$$

Consider an arbitrary unitary depending on a single phase-space coordinate,

$$U(\theta, \hat{x}) = e^{-i\theta f(\hat{x})}, \quad (\text{A2})$$

acting on this generic pure Gaussian state. The resulting Wigner function takes the form

$$W(x, p; \theta) = \frac{1}{2\pi} \sqrt{\frac{2a}{\pi}} e^{-2a(x-x_0)^2} \times \int dy e^{-\frac{a}{2}y^2} e^{iy[p-p_0+2b(x-x_0)]} e^{i\theta[f(x+y/2)-f(x-y/2)]}. \quad (\text{A3})$$

The Wigner logarithmic negativity is determined by integrating the absolute value of the Wigner function over phase-space,

$$e^{\mathcal{M}} = \int dx dp |W(x, p; \theta)| = \theta \int dx dv |W(x, \theta v; \theta)|, \quad (\text{A4})$$

where we have introduced the rescaled momentum variable $p = \theta v$. Under this change of variables, the integrand maps to a slowly varying amplitude multiplying a highly oscillatory phase driven by θ

$$W(x, \theta v; \theta) = \int_{-\infty}^{\infty} dy g(y, x) e^{i\theta \Phi(y, x, v)}, \quad (\text{A5})$$

The dominant phase scaling with θ is governed by

$$\Phi(y, x, v) = vy + f(x + y/2) - f(x - y/2). \quad (\text{A6})$$

while the remaining θ -independent terms, including the Gaussian envelope and initial state displacements, are absorbed into the complex amplitude function $g(y, x)$. As $\theta \rightarrow \infty$, the highly oscillatory nature of the integrand restricts the dominant contributions to the stationary points y^* satisfying $\partial\Phi/\partial y = 0$. Assuming a single dominant non-degenerate stationary point y_0^* where $\Phi''(y_0^*) \neq 0$, the integral evaluates to

$$W(x, \theta v; \theta) \approx \sqrt{\frac{2\pi}{\theta |\Phi''(y_0^*)|}} g(y_0^*, x) e^{i(\theta \Phi(y_0^*) + \frac{\pi}{4} \text{sgn}(\Phi''(y_0^*)))}. \quad (\text{A7})$$

The condition $\Phi''(y^*) = 0$ can occur for non-Gaussian generators on specific lower-dimensional manifolds (caustics), such as the curve $y^* = 0$ corresponding to $v = -f'(x)$. However, because these caustic regions occupy a set of measure zero in the

full 2D phase-space integration, their contribution to the global scaling is subdominant as $\theta \rightarrow \infty$.

Consequently, the bulk of the phase-space integration yields an integrand scaling as $\theta^{-1/2}$. Combined with the factor of θ originating from the integration measure change $dp = \theta dv$, the total scaling of the integral behaves as $\theta/\sqrt{\theta} = \sqrt{\theta}$. This establishes the universal asymptotic growth behavior of the logarithmic negativity

$$\mathcal{M} \xrightarrow{\theta \rightarrow \infty} \frac{1}{2} \ln \theta + \dots \quad (\text{A8})$$

If multiple isolated stationary points exist, this scaling law remains valid. The absolute value inside the logarithmic negativity causes the highly oscillatory interference cross-terms between different stationary points to average to a geometric constant independent of θ upon global integration.

Appendix B: Asymptotic approximation

We consider the operator $\hat{\mathcal{C}}_1(\theta, c) = \hat{G}_1 \otimes \mathbb{I} + \hat{G}_2 \otimes Z + \hat{B} \otimes X$ as in Eq. (36). Given that all the bosonic operators commute with each other, the n -th power is found to have the same structure $\hat{\mathcal{C}}_1^n = A_n \otimes \mathbb{I} + Z_n \otimes Z + X_n \otimes X$. The coefficients follow the recurrence relation

$$\begin{pmatrix} A_{n+1} \\ Z_{n+1} \\ X_{n+1} \end{pmatrix} = \begin{pmatrix} \hat{G}_1 & \hat{G}_2 & \hat{B} \\ \hat{G}_2 & \hat{G}_1 & 0 \\ \hat{B} & 0 & \hat{G}_1 \end{pmatrix} \begin{pmatrix} A_n \\ Z_n \\ X_n \end{pmatrix}. \quad (\text{B1})$$

Since $\hat{G}_2$ and $\hat{B}$ are anti-Hermitian, their squares are negative semi-definite. We define the positive semi-definite Hermitian operator $\hat{R} = \sqrt{-(\hat{G}_2^2 + \hat{B}^2)}$. Diagonalizing the coefficient matrix yields the closed-form coefficients

$$A_n = \frac{1}{2} [(\hat{G}_1 + i\hat{R})^n + (\hat{G}_1 - i\hat{R})^n], \quad (\text{B2})$$

$$Z_n = \frac{\hat{G}_2}{2i\hat{R}} [(\hat{G}_1 + i\hat{R})^n - (\hat{G}_1 - i\hat{R})^n], \quad (\text{B3})$$

$$X_n = \frac{\hat{B}}{2i\hat{R}} [(\hat{G}_1 + i\hat{R})^n - (\hat{G}_1 - i\hat{R})^n]. \quad (\text{B4})$$

To verify the limit $\theta \mapsto \theta/n$ as $n \rightarrow \infty$, we note $\hat{G}_1 \rightarrow \mathbb{I}$, $\hat{G}_2 \rightarrow -i\frac{\theta}{n} \cos(c\hat{x})$, $\hat{B} \rightarrow 0$, and thus $\hat{R} \rightarrow$

$\frac{\theta}{n} \cos(c\hat{x})$. Therefore, we get

$$\begin{aligned} A_n &\rightarrow \frac{1}{2} \left[\left(1 + i \frac{\theta}{n} \cos(c\hat{x}) \right)^n + \left(1 - i \frac{\theta}{n} \cos(c\hat{x}) \right)^n \right] \\ &\rightarrow \cos(\theta \cos(c\hat{x})), \\ Z_n &\rightarrow \frac{-1}{2} \left[\left(1 + i \frac{\theta}{n} \cos(c\hat{x}) \right)^n - \left(1 - i \frac{\theta}{n} \cos(c\hat{x}) \right)^n \right] \\ &\rightarrow -i \sin(\theta \cos(c\hat{x})), \\ X_n &\rightarrow 0. \end{aligned} \quad (\text{B5})$$

Consequently, the operator converges to the trigonometric gate

$$\begin{aligned} \hat{C}_1^n &\rightarrow \cos(\theta \cos(c\hat{x})) \otimes \mathbb{I} - i \sin(\theta \cos(c\hat{x})) \otimes Z \\ &= \exp(-i\theta \cos(c\hat{x}) \otimes Z). \end{aligned} \quad (\text{B6})$$

Appendix C: Equivalence of the ancilla constructions

Here, we briefly show how the two-ancilla gate qubit construction of the trigonometric gates presented in Ref. [63] is related to the single-ancilla construction of Ref. [64]. The two-ancilla construction is based on the following structure of hybrid qubit-quumode unitaries

$$\begin{aligned} \Sigma_{ij} &\equiv \cos(\hat{A}) \otimes \sigma_i + \sin(\hat{A}) \otimes \sigma_j, \\ \bar{\Sigma}_{ij} &\equiv \cos(\hat{A}) \otimes \sigma_i - \sin(\hat{A}) \otimes \sigma_j, \end{aligned} \quad (\text{C1})$$

that can be exponentiated by exploiting a qubit-ancilla based method to approximate continuous-variable trigonometric gates of the form

$$\begin{aligned} e^{-i\frac{\theta}{2}\Sigma_{ij}} e^{-i\frac{\theta}{2}\bar{\Sigma}_{ij}} &= e^{-i\theta \cos(\hat{A}) \otimes \sigma_i} + \mathcal{O}(\theta^2), \\ e^{-i\frac{\theta}{2}\Sigma_{ij}} e^{i\frac{\theta}{2}\bar{\Sigma}_{ij}} &= e^{-i\theta \sin(\hat{A}) \otimes \sigma_j} + \mathcal{O}(\theta^2), \end{aligned} \quad (\text{C2})$$

with $\hat{A}$ any Hermitian operator. The method described in Ref. [63] exploits two ancilla gate qubits for the trigonometric gates construction. However, the final circuit can be unitarily reduced to requiring solely a single ancilla gate qubit [64]. Without loss of generality, we work with $\Sigma_{zy} \equiv \Sigma$, which corresponds to the same choice of both Refs. [63, 64]. The ancilla-based exponentiation of Ref. [63] exactly provides the hybrid gate $e^{-i\theta \Sigma \otimes Z_b}$, as opposed to simply $e^{-i\theta \Sigma \otimes \mathbb{I}_b}$. The latter being created using the circuit in Ref. [64]. It is clear that there exists a unitary transformation V such that

$$V e^{-i\theta \Sigma \otimes Z_b} V^\dagger = e^{-i\theta \Sigma \otimes \mathbb{I}_b}. \quad (\text{C3})$$

For our choice, it is $V = \text{CNOT}_{b \rightarrow a}$, where a labels the ancilla gate qubit in Σ . Note that this is an operator identity. The secondary gate qubit b decouples exactly, and it is not projected onto its eigenstate.

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- [1] S. Lloyd and S. L. Braunstein, “Quantum computation over continuous variables,” *Phys. Rev. Lett.* **82** (1999) 1784–1787, [arXiv:quant-ph/9810082](#).
 - [2] S. L. Braunstein and P. van Loock, “Quantum information with continuous variables,” *Rev. Mod. Phys.* **77** (Jun, 2005) 513–577. <https://link.aps.org/doi/10.1103/RevModPhys.77.513>.
 - [3] U. L. Andersen, J. S. Neergaard-Nielsen, P. van Loock, and A. Furusawa, “Hybrid discrete- and continuous-variable quantum information,” *Nature Physics* **11** no. 9, (2015) 713–719. <https://doi.org/10.1038/nphys3410>.
 - [4] D. Leibfried, R. Blatt, C. Monroe, and D. Wineland, “Quantum dynamics of single trapped ions,” *Rev. Mod. Phys.* **75** (Mar, 2003) 281–324. <https://link.aps.org/doi/10.1103/RevModPhys.75.281>.
 - [5] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, “Circuit quantum electrodynamics,” *Rev. Mod. Phys.* **93** no. 2, (2021) 025005, [arXiv:2005.12667 \[quant-ph\]](#).
 - [6] T. J. Stavenger, E. Crane, K. C. Smith, C. T. Kang, S. M. Girvin, and N. Wiebe, “C2QA - Bosonic Qiskit,” in *26th IEEE High Performance Extreme Computing*, 9, 2022. [arXiv:2209.11153 \[quant-ph\]](#).
 - [7] Y. Liu, S. Singh, K. C. Smith, E. Crane, J. M. Martyn, A. Eickbusch, A. Schuckert, R. D. Li, J. Sinanan-Singh, M. B. Soley, T. Tsunoda, I. L. Chuang, N. Wiebe, and S. M. Girvin, “Hybrid oscillator-qubit quantum processors: Instruction set architectures, abstract machine models, and applications,” *PRX Quantum* **7** (Jan, 2026) 010201. <https://link.aps.org/doi/10.1103/4rf7-9tfx>.
 - [8] J. Y. Araz, M. Grau, J. Montgomery, and F. Ringer, “Hybrid quantum simulations with

- qubits and qumodes on trapped-ion platforms,” *Phys. Rev. A* **112** no. 1, (2025) 012620, [arXiv:2410.07346 \[quant-ph\]](#).
- [9] C. Flühmann, T. L. Nguyen, M. Marinelli, V. Negnevitsky, K. Mehta, and J. Home, “Encoding a qubit in a trapped-ion mechanical oscillator,” *Nature* **566** no. 7745, (2019) 513–517, [arXiv:1807.01033 \[quant-ph\]](#).
- [10] B. de Neeve, T.-L. Nguyen, T. Behrle, and J. P. Home, “Error correction of a logical grid state qubit by dissipative pumping,” *Nature Physics* **18** no. 3, (2022) 296–300. [https://doi.org/10.1038/s41567-021-01487-7](#).
- [11] R. W. Heeres, B. Vlastakis, E. Holland, S. Krastanov, V. V. Albert, L. Frunzio, L. Jiang, and R. J. Schoelkopf, “Cavity State Manipulation Using Photon-Number Selective Phase Gates,” *Phys. Rev. Lett.* **115** no. 13, (2015) 137002, [arXiv:1503.01496 \[quant-ph\]](#).
- [12] A. Eickbusch, V. Sivak, A. Z. Ding, S. S. Elder, S. R. Jha, J. Venkatraman, B. Royer, S. M. Girvin, R. J. Schoelkopf, and M. H. Devoret, “Fast universal control of an oscillator with weak dispersive coupling to a qubit,” *Nature Phys.* **18** (2022) 1464–1469, [arXiv:2111.06414 \[quant-ph\]](#).
- [13] C. S. Wang, J. C. Curtis, B. J. Lester, Y. Zhang, Y. Y. Gao, J. Freeze, V. S. Batista, P. H. Vaccaro, I. L. Chuang, L. Frunzio, L. Jiang, S. M. Girvin, and R. J. Schoelkopf, “Efficient multiphoton sampling of molecular vibronic spectra on a superconducting bosonic processor,” *Phys. Rev. X* **10** no. 2, (2020) 021060, [arXiv:1908.03598 \[quant-ph\]](#).
- [14] P. C. Haljan, K.-A. Brickman, L. Deslauriers, P. J. Lee, and C. Monroe, “Spin-dependent forces on trapped ions for phase-stable quantum gates and entangled states of spin and motion,” *Phys. Rev. Lett.* **94** (Apr, 2005) 153602. [https://link.aps.org/doi/10.1103/PhysRevLett.94.153602](#).
- [15] W. Chen, Y. Lu, S. Zhang, K. Zhang, G. Huang, M. Qiao, X. Su, J. Zhang, J.-N. Zhang, L. Bianchi, M. S. Kim, and K. Kim, “Scalable and programmable phononic network with trapped ions,” *Nature Physics* **19** no. 6, (2023) 877–883. [https://doi.org/10.1038/s41567-023-01952-5](#).
- [16] D. M. Meekhof, C. Monroe, B. E. King, W. M. Itano, and D. J. Wineland, “Generation of nonclassical motional states of a trapped atom,” *Phys. Rev. Lett.* **76** (Mar, 1996) 1796–1799. [https://link.aps.org/doi/10.1103/PhysRevLett.76.1796](#).
- [17] M. Um, J. Zhang, D. Lv, Y. Lu, S. An, J.-N. Zhang, H. Nha, M. S. Kim, and K. Kim, “Phonon arithmetic in a trapped ion system,” *Nature Communications* **7** no. 1, (2016) 11410. [https://doi.org/10.1038/ncomms11410](#).
- [18] S. Mohapatra, Y. Liu, E. Z. Zhang, and H. Zhou, “HyQBench: A Benchmark Suite for Hybrid CV-DV Quantum Computing,” [arXiv:2603.04398 \[quant-ph\]](#).
- [19] M. H. Heris, Y. Liu, and F. Mueller, “HyPulse: A Pulse Synthesis Framework for Hybrid Qubit-Oscillator Gates on Trapped-Ion Platform,” [arXiv:2604.26804 \[quant-ph\]](#).
- [20] R. T. Sutherland and R. Srinivas, “Universal hybrid quantum computing in trapped ions,” *Phys. Rev. A* **104** no. 3, (2021) 032609, [arXiv:2105.05768 \[quant-ph\]](#).
- [21] W. Chen, J. Gan, J.-N. Zhang, D. Matuskevich, and K. Kim, “Quantum computation and simulation with vibrational modes of trapped ions,” *Chinese Physics B* **30** no. 6, (Jun, 2021) 060311. [https://doi.org/10.1088/2F1674-1056/2Fac01e3](#).
- [22] A. F. Kemper, A. Alvertis, M. Asaduzzaman, B. N. Bakalov, D. Baron, J. Bierman, B. Burgstahler, S. Chundury, E. R. Das, J. Furches, F. Guo, R. G. Jha, K. Klymko, A. Kushwaha, A. Li, A. Majumdar, C. O. Marrero, S. Mohapatra, C. Mori, F. Mueller, D. T. Popovici, T. Stavenger, M. Tirfe, N. M. Tubman, M. Zheng, H. Zhou, and Y. Liu, “Hybrid continuous-discrete-variable quantum computing: a guide to utility,” 2025. [https://arxiv.org/abs/2511.13882](#).
- [23] D. Gottesman, A. Kitaev, and J. Preskill, “Encoding a qubit in an oscillator,” *Phys. Rev. A* **64** (2001) 012310, [arXiv:quant-ph/0008040](#).
- [24] L. Hu, Y. Ma, W. Cai, X. Mu, Y. Xu, W. Wang, Y. Wu, H. Wang, Y. P. Song, C. L. Zou, S. M. Girvin, L.-M. Duan, and L. Sun, “Quantum error correction and universal gate set operation on a binomial bosonic logical qubit,” *Nature Physics* **15** no. 5, (2019) 503–508. [https://doi.org/10.1038/s41567-018-0414-3](#).
- [25] P. Campagne-Ibarcq, A. Eickbusch, S. Touzard, E. Zolys-Geller, N. E. Frattini, V. V. Sivak, P. Reinhold, S. Puri, S. Shankar, R. J. Schoelkopf, L. Frunzio, M. Mirrahimi, and M. H. Devoret, “Quantum error correction of a qubit encoded in grid states of an oscillator,” *Nature* **584** no. 7821, (2020) 368–372. [https://doi.org/10.1038/s41586-020-2603-3](#).

- [26] B. L. Brock, S. Singh, A. Eickbusch, V. V. Sivak, A. Z. Ding, L. Frunzio, S. M. Girvin, and M. H. Devoret, “Quantum error correction of qudits beyond break-even,” *Nature* **641** no. 8063, (2025) 612–618, [arXiv:2409.15065 \[quant-ph\]](#).
- [27] V. G. Matsos, C. H. Valahu, T. Navickas, A. D. Rao, M. J. Millican, X. C. Kolesnikow, M. J. Biercuk, and T. R. Tan, “Robust and deterministic preparation of bosonic logical states in a trapped ion,” *Phys. Rev. Lett.* **133** (Jul, 2024) 050602. <https://link.aps.org/doi/10.1103/PhysRevLett.133.050602>.
- [28] V. G. Matsos, C. H. Valahu, M. J. Millican, T. Navickas, X. C. Kolesnikow, M. J. Biercuk, and T. R. Tan, “Universal quantum gate set for Gottesman–Kitaev–Preskill logical qubits,” *Nature Phys.* **21** no. 10, (2025) 1664–1669, [arXiv:2409.05455 \[quant-ph\]](#).
- [29] P. Jordan and E. P. Wigner, “About the Pauli exclusion principle,” *Z. Phys.* **47** (1928) 631–651.
- [30] E. Chiari, W. Makhlof, L. Pepe, E. Koridon, J. Klein, B. Senjean, B. Lasorne, and S. Yalouz, “Ab initio polaritonic chemistry on diverse quantum computing platforms: Ansatz circuit design for qubit, qudit, and hybrid qubit-qumode architectures,” *Phys. Rev. A* **112** no. 5, (2025) 052433, [arXiv:2506.12504 \[quant-ph\]](#).
- [31] S. Debnath, N. M. Linke, S.-T. Wang, C. Figgatt, K. A. Landsman, L.-M. Duan, and C. Monroe, “Observation of hopping and blockade of bosons in a trapped ion spin chain,” *Phys. Rev. Lett.* **120** (Feb, 2018) 073001. <https://link.aps.org/doi/10.1103/PhysRevLett.120.073001>.
- [32] J. B. Kogut and L. Susskind, “Hamiltonian Formulation of Wilson’s Lattice Gauge Theories,” *Phys. Rev. D* **11** (1975) 395–408.
- [33] S. P. Jordan, K. S. M. Lee, and J. Preskill, “Quantum Computation of Scattering in Scalar Quantum Field Theories,” *Quant. Inf. Comput.* **14** (2014) 1014–1080, [arXiv:1112.4833 \[hep-th\]](#).
- [34] E. Zohar, J. I. Cirac, and B. Reznik, “Quantum Simulations of Lattice Gauge Theories using Ultracold Atoms in Optical Lattices,” *Rept. Prog. Phys.* **79** no. 1, (2016) 014401, [arXiv:1503.02312 \[quant-ph\]](#).
- [35] D. Banerjee, M. Dalmonte, M. Muller, E. Rico, P. Stebler, U. J. Wiese, and P. Zoller, “Atomic Quantum Simulation of Dynamical Gauge Fields coupled to Fermionic Matter: From String Breaking to Evolution after a Quench,” *Phys. Rev. Lett.* **109** (2012) 175302, [arXiv:1205.6366 \[cond-mat.quant-gas\]](#).
- [36] M. C. Bañuls, R. Blatt, J. Catani, A. Celi, J. I. Cirac, M. Dalmonte, L. Fallani, K. Jansen, M. Lewenstein, S. Montangero, C. A. Muschik, B. Reznik, E. Rico, L. Tagliacozzo, K. Van Acoleyen, F. Verstraete, U.-J. Wiese, M. Wingate, J. Zakrzewski, and P. Zoller, “Simulating lattice gauge theories within quantum technologies,” *The European Physical Journal D* **74** no. 8, (2020) 165. <https://doi.org/10.1140/epjd/e2020-100571-8>.
- [37] M. Kreshchuk, S. Jia, W. M. Kirby, G. Goldstein, J. P. Vary, and P. J. Love, “Simulating Hadronic Physics on NISQ devices using Basis Light-Front Quantization,” *Phys. Rev. A* **103** no. 6, (2021) 062601, [arXiv:2011.13443 \[quant-ph\]](#).
- [38] J. F. Haase, L. Dellantonio, A. Celi, D. Paulson, A. Kan, K. Jansen, and C. A. Muschik, “A resource efficient approach for quantum and classical simulations of gauge theories in particle physics,” *Quantum* **5** (2021) 393, [arXiv:2006.14160 \[quant-ph\]](#).
- [39] E. Zohar, “Quantum simulation of lattice gauge theories in more than one space dimension—requirements, challenges and methods,” *Phil. Trans. A. Math. Phys. Eng. Sci.* **380** no. 2216, (2022) 20210069, [arXiv:2106.04609 \[quant-ph\]](#).
- [40] N. Klco and M. J. Savage, “Digitization of scalar fields for quantum computing,” *Phys. Rev. A* **99** no. 5, (2019) 052335, [arXiv:1808.10378 \[quant-ph\]](#).
- [41] C. W. Bauer, Z. Davoudi, A. B. Balantekin, T. Bhattacharya, M. Carena, W. A. de Jong, P. Draper, A. El-Khadra, N. Gemelke, M. Hanada, D. Kharzeev, H. Lamm, Y.-Y. Li, J. Liu, M. Lukin, Y. Meurice, C. Monroe, B. Nachman, G. Pagano, J. Preskill, E. Rinaldi, A. Roggero, D. I. Santiago, M. J. Savage, I. Siddiqi, G. Siopsis, D. Van Zanten, N. Wiebe, Y. Yamauchi, K. Yeter-Aydeniz, and S. Zorzetti, “Quantum simulation for high-energy physics,” *PRX Quantum* **4** (May, 2023) 027001. <https://link.aps.org/doi/10.1103/PRXQuantum.4.027001>.
- [42] R. A. Briceño, R. G. Edwards, M. Eaton, C. González-Arciniegas, O. Pfister, and G. Siopsis, “Toward coherent quantum computation of scattering amplitudes with a measurement-based photonic quantum processor,” *Phys. Rev. Res.* **6** no. 4, (2024) 043065, [arXiv:2312.12613 \[quant-ph\]](#).
- [43] Z. Davoudi, N. M. Linke, and G. Pagano, “Toward simulating quantum field theories with controlled

- phonon-ion dynamics: A hybrid analog-digital approach,” *Phys. Rev. Res.* **3** no. 4, (2021) 043072, [arXiv:2104.09346 \[quant-ph\]](#).
- [44] O. Katz and C. Monroe, “Programmable Quantum Simulations of Bosonic Systems with Trapped Ions,” *Phys. Rev. Lett.* **131** no. 3, (2023) 033604, [arXiv:2207.13653 \[quant-ph\]](#).
- [45] E. J. Gustafson, Y. Ji, H. Lamm, E. M. Murairi, S. O. Perez, and S. Zhu, “Primitive quantum gates for an $SU(3)$ discrete subgroup: $\Sigma(36\times 3)$,” *Phys. Rev. D* **110** no. 3, (2024) 034515, [arXiv:2405.05973 \[hep-lat\]](#).
- [46] R. C. Farrell, M. Illa, A. N. Ciavarella, and M. J. Savage, “Scalable Circuits for Preparing Ground States on Digital Quantum Computers: The Schwinger Model Vacuum on 100 Qubits,” *PRX Quantum* **5** no. 2, (2024) 020315, [arXiv:2308.04481 \[quant-ph\]](#).
- [47] V. Ale, N. M. Bauer, R. G. Jha, F. Ringer, and G. Siopsis, “Quantum computation of $SU(2)$ lattice gauge theory with continuous variables,” *JHEP* **06** (2025) 084, [arXiv:2410.14580 \[hep-lat\]](#).
- [48] S. Abel, M. Spannowsky, and S. Williams, “Real-time scattering processes with continuous-variable quantum computers,” *Phys. Rev. A* **112** no. 1, (2025) 012614, [arXiv:2502.01767 \[quant-ph\]](#).
- [49] E. Crane, K. C. Smith, T. Tomesh, A. Eickbusch, J. M. Martyn, S. Kühn, L. Funcke, M. A. DeMarco, I. L. Chuang, N. Wiebe, A. Schuckert, and S. M. Girvin, “Hybrid oscillator-qubit quantum processors: Simulating fermions, bosons, and gauge fields,” 9, 2024.
- [50] V. Ale, T. Rainaldi, E. Rico, F. Ringer, and G. Siopsis, “Simulating quantum electrodynamics in 2+1 dimensions with qubits and qumodes,” *JHEP* **04** (2026) 122, [arXiv:2511.14506 \[quant-ph\]](#).
- [51] S. Saner, O. Băzăvan, D. J. Webb, G. Araneda, C. J. Ballance, R. Srinivas, D. M. Lucas, and A. Bermúdez, “Real-Time Observation of Aharonov-Bohm Interference in a $\mathbb{Z}_2$ Lattice Gauge Theory on a Hybrid Qubit-Oscillator Quantum Computer,” [arXiv:2507.19588 \[quant-ph\]](#).
- [52] A. T. Than, S. V. Kadam, V. Vikramaditya, N. H. Nguyen, X. Liu, Z. Davoudi, A. M. Green, and N. M. Linke, “Observation of quantum-field-theory dynamics on a spin-phonon quantum computer,” 2025. <https://arxiv.org/abs/2509.11477>.
- [53] D. Zou, T. Li, J. Liang, E. Wang, and H. Xing, “Hadronic tensor in lattice gauge theories by quantum computing,” [arXiv:2606.17003 \[hep-ph\]](#).
- [54] D. Athanasakos, G. Tejedor-García, J. Y. Araz, M. Ramôa, B. Sambasivam, S. E. Economou, and F. Ringer, “Continuous-variable ADAPT-VQE for bosonic lattice models,” [arXiv:2606.05297 \[quant-ph\]](#).
- [55] C. V. Cogburn, S. Grieninger, and D. E. Kharzeev, “Quantum Simulation of Nucleon-Antinucleon Interaction in Large- N QCD₂ on an IBM Quantum Nighthawk Processor,” [arXiv:2606.02574 \[quant-ph\]](#).
- [56] D. Lv, S. An, Z. Liu, J.-N. Zhang, J. S. Pedernales, L. Lamata, E. Solano, and K. Kim, “Quantum simulation of the quantum rabi model in a trapped ion,” *Phys. Rev. X* **8** (Apr, 2018) 021027. <https://link.aps.org/doi/10.1103/PhysRevX.8.021027>.
- [57] S. Hahn and G. Stock, “Quantum-mechanical modeling of the femtosecond isomerization in rhodopsin,” *The Journal of Physical Chemistry B* **104** no. 6, (2000) 1146–1149, <https://doi.org/10.1021/jp992939g>. <https://doi.org/10.1021/jp992939g>.
- [58] Y. Shen, Y. Lu, K. Zhang, J. Zhang, S. Zhang, J. Huh, and K. Kim, “Quantum optical emulation of molecular vibronic spectroscopy using a trapped-ion device,” *Chem. Sci.* **9** (2018) 836–840. <http://dx.doi.org/10.1039/C7SC04602B>.
- [59] C. S. Wang, J. C. Curtis, B. J. Lester, Y. Zhang, Y. Y. Gao, J. Freeze, V. S. Batista, P. H. Vaccaro, I. L. Chuang, L. Frunzio, L. Jiang, S. M. Girvin, and R. J. Schoelkopf, “Efficient multiphoton sampling of molecular vibronic spectra on a superconducting bosonic processor,” *Phys. Rev. X* **10** (Jun, 2020) 021060. <https://link.aps.org/doi/10.1103/PhysRevX.10.021060>.
- [60] C. H. Valahu, V. C. Olaya-Agudelo, R. J. MacDonell, T. Navickas, A. D. Rao, M. J. Millican, J. B. Pérez-Sánchez, J. Yuen-Zhou, M. J. Biercuk, C. Hempel, T. R. Tan, and I. Kassal, “Direct observation of geometric-phase interference in dynamics around a conical intersection,” *Nature Chemistry* **15** no. 11, (2023) 1503–1508. <https://doi.org/10.1038/s41557-023-01300-3>.
- [61] R. Dutta, D. G. A. Cabral, N. Lyu, N. P. Vu, Y. Wang, B. Allen, X. Dan, R. G. Cortiñas, P. Khazaei, M. Schäfer, A. C. C. d. Albornoz, S. E. Smart, S. Nie, M. H. Devoret, D. A. Mazziotti, P. Narang, C. Wang, J. D. Whitfield, A. K. Wilson, H. P. Hendrickson, D. A. Lidar, F. Pérez-Bernal, L. F. Santos, S. Kais, E. Geva, and V. S. Batista, “Simulating chemistry on bosonic quantum

- devices,” *Journal of Chemical Theory and Computation* **20** no. 15, (08, 2024) 6426–6441. <https://doi.org/10.1021/acs.jctc.4c00544>.
- [62] S. R. Coleman, “The Quantum Sine-Gordon Equation as the Massive Thirring Model,” *Phys. Rev. D* **11** (1975) 2088.
- [63] T. Rainaldi, V. Ale, M. Grau, D. Kharzeev, E. Rico, F. Ringer, P. Shome, and G. Siopsis, “Trigonometric continuous-variable gates and hybrid quantum simulations of the sine-Gordon model,” *JHEP* **03** (2026) 125, [arXiv:2512.19582](https://arxiv.org/abs/2512.19582) [quant-ph].
- [64] T. Chalermputitarak, K. Schwennicke, I. Kassal, and T. R. Tan, “Programmable generation of arbitrary continuous-variable anharmonicities and nonlinear couplings,” [arXiv:2511.22286](https://arxiv.org/abs/2511.22286) [quant-ph].
- [65] S. M. Clark, D. Lobser, M. C. Revelle, C. G. Yale, D. Bossert, A. D. Burch, M. N. Chow, C. W. Hogle, M. Ivory, J. Pehr, B. Salzbrenner, D. Stick, W. Sweatt, J. M. Wilson, E. Winrow, and P. Maunz, “Engineering the quantum scientific computing open user testbed,” *IEEE Transactions on Quantum Engineering* **2** (2021) 1–32.
- [66] Z. Jia, Y. Wang, B. Zhang, J. Whitlow, C. Fang, J. Kim, and K. R. Brown, “Determination of multimode motional quantum states in a trapped ion system,” *Phys. Rev. Lett.* **129** (Aug, 2022) 103602. <https://link.aps.org/doi/10.1103/PhysRevLett.129.103602>.
- [67] C. McGarry, T. Chalermputitarak, K. Schwennicke, F. Scuccimarra, M. J. Millican, V. G. Matsos, C. H. Valahu, P. Nagpal, H.-K. Chan, H. L. Nourse, I. Kassal, and T. R. Tan, “Programmable quantum simulation of anharmonic dynamics,” 2026. <https://arxiv.org/abs/2603.04744>.
- [68] O. Băzăvan, S. Saner, D. J. Webb, E. M. Ainley, P. Drmota, D. P. Nadlinger, G. Araneda, D. M. Lucas, C. J. Ballance, and R. Srinivas, “Squeezing, trisqueezing and quadsqueezing in a hybrid oscillator–spin system,” *Nature Physics* **22** no. 5, (2026) 757–762. <https://doi.org/10.1038/s41567-026-03222-6>.
- [69] J. Sinanan-Singh, G. L. Mintzer, I. L. Chuang, and Y. Liu, “Single-shot Quantum Signal Processing Interferometry,” *Quantum* **8** (2024) 1427, [arXiv:2311.13703](https://arxiv.org/abs/2311.13703) [quant-ph].
- [70] Y. Liu, J. M. Martyn, J. Sinanan-Singh, K. C. Smith, S. M. Girvin, and I. L. Chuang, “Toward mixed analog-digital quantum signal processing: Quantum ad/da conversion and the fourier transform,” *IEEE Transactions on Signal Processing* **73** (2025) 3641–3655.
- [71] J. Hong, S. H. Kim, S. K. Min, and J. Huh, “Oscillator-qubit generalized quantum signal processing for vibronic models: a case study of uracil cation,” 2025. <https://arxiv.org/abs/2510.10495>.
- [72] E. C. Tortorici, E. C. McGarrigle, B. K. McFarland, W. L. Johnson, D. S. Lobser, M. C. Revelle, B. P. Ruzic, S. M. Clark, and C. G. Yale, “QSCOUT’s Qubit-Boson Gate Set,” [arXiv:2607.08560](https://arxiv.org/abs/2607.08560) [quant-ph].
- [73] M. C. Revelle, “Phoenix and peregrine ion traps,” 2020. <https://arxiv.org/abs/2009.02398>.
- [74] D. J. Wineland, C. Monroe, W. M. Itano, D. Leibfried, B. E. King, and D. M. Meekhof, “Experimental issues in coherent quantum-state manipulation of trapped atomic ions,” 1998. <https://arxiv.org/abs/quant-ph/9710025>.
- [75] C. Flühmann, *Encoding a qubit in the motion of a trapped ion using superpositions of displaced squeezed states*. PhD thesis, ETH Zürich, 2019. <https://doi.org/10.3929/ethz-b-000355836>.
- [76] C. Flühmann and J. P. Home, “Direct characteristic-function tomography of quantum states of the trapped-ion motional oscillator,” *Phys. Rev. Lett.* **125** (Jul, 2020) 043602. <https://link.aps.org/doi/10.1103/PhysRevLett.125.043602>.
- [77] D. Leibfried, D. M. Meekhof, B. E. King, C. Monroe, W. M. Itano, and D. J. Wineland, “Experimental determination of the motional quantum state of a trapped atom,” *Phys. Rev. Lett.* **77** (Nov, 1996) 4281–4285. <https://link.aps.org/doi/10.1103/PhysRevLett.77.4281>.
- [78] D. M. Meekhof, C. Monroe, B. E. King, W. M. Itano, and D. J. Wineland, “Generation of nonclassical motional states of a trapped atom,” *Phys. Rev. Lett.* **76** (Mar, 1996) 1796–1799. <https://link.aps.org/doi/10.1103/PhysRevLett.76.1796>.
- [79] D. Lv, S. An, M. Um, J. Zhang, J.-N. Zhang, M. S. Kim, and K. Kim, “Reconstruction of the jaynes-cummings field state of ionic motion in a harmonic trap,” *Phys. Rev. A* **95** (Apr, 2017) 043813. <https://link.aps.org/doi/10.1103/PhysRevA.95.043813>.
- [80] K. E. Cahill and R. J. Glauber, “Ordered expansions in boson amplitude operators,” *Phys. Rev.* **177** (1969) 1857–1881.
- [81] F. Albarelli, M. G. Genoni, M. G. A. Paris, and A. Ferraro, “Resource theory of quantum

- non-Gaussianity and Wigner negativity,” *Phys. Rev. A* **98** no. 5, (2018) 052350, [arXiv:1804.05763 \[quant-ph\]](#).
- [82] M. Suzuki, “Fractal decomposition of exponential operators with applications to many-body theories and monte carlo simulations,” *Physics Letters A* **146** no. 6, (1990) 319–323. <https://www.sciencedirect.com/science/article/pii/037596019090962N>.
- [83] N. Wiebe, D. W. Berry, P. Høyer, and B. C. Sanders, “Higher order decompositions of ordered operator exponentials,” *J. Phys. A* **43** no. 6, (2010) 065203, [arXiv:0812.0562 \[math-ph\]](#).
- [84] A. Mercurio, Y.-T. Huang, L.-X. Cai, Y.-N. Chen, V. Savona, and F. Nori, “QuantumToolbox.jl: An efficient Julia framework for simulating open quantum systems,” *Quantum* **9** (Sept., 2025) 1866. <https://doi.org/10.22331/q-2025-09-29-1866>.
- [85] J. Bezanson, A. Edelman, S. Karpinski, and V. B. Shah, “Julia: A fresh approach to numerical computing,” *SIAM Review* **59** no. 1, (2017) 65–98. <https://epubs.siam.org/doi/10.1137/141000671>.